Basic 10X Genomics Chromium data analysis

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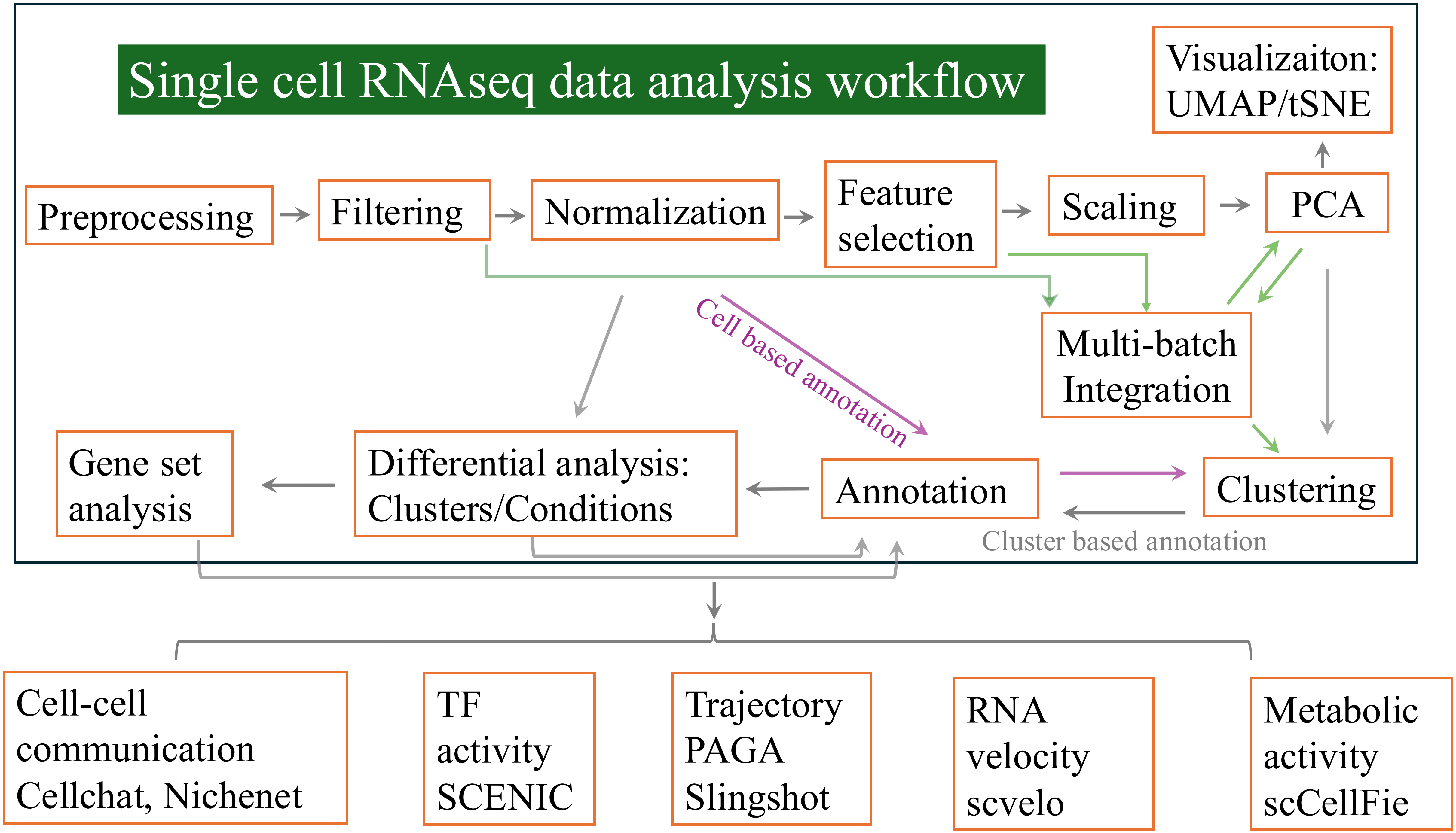
NBIS
Immunotechology

Learnin goals

- Understand why filter data
- Understand why normalize data
- Understand why scale data?
- Understand why do dimensional reduction
- Know the popular dimensional reduction methods
- Understand why do clustering
- Understand why do differential analysis
- Understand why do cell annotation


Single cell RNAseq data analysis workflow
Visualizaiton:
UMAP/tSNE
Preprocessing
Filtering
Normalization
Feature
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Scaling
PCA
Multi-batch
Integration
Cell based annotation
Gene set
analysis
Differential analysis:
Clusters/Conditions
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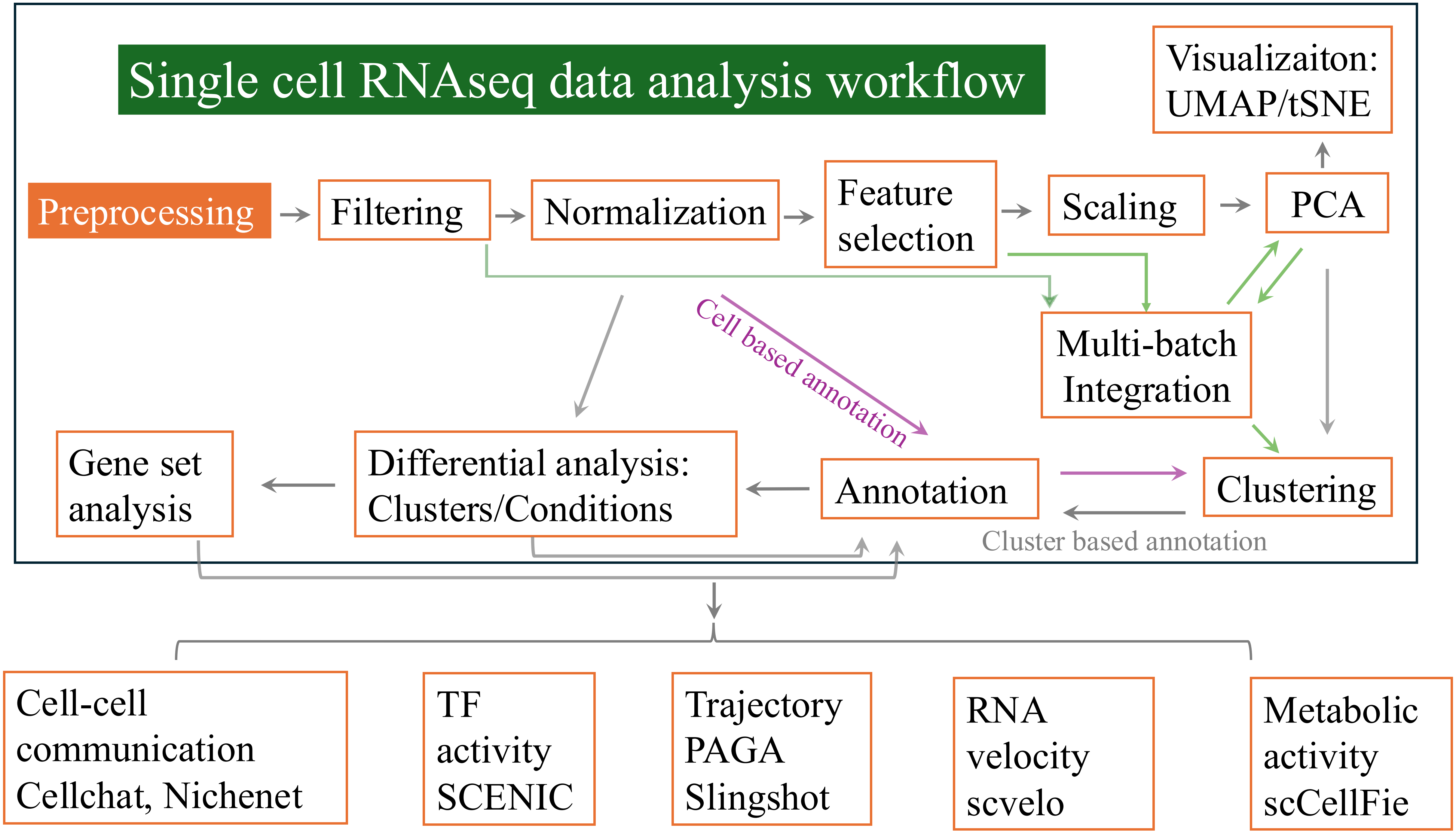
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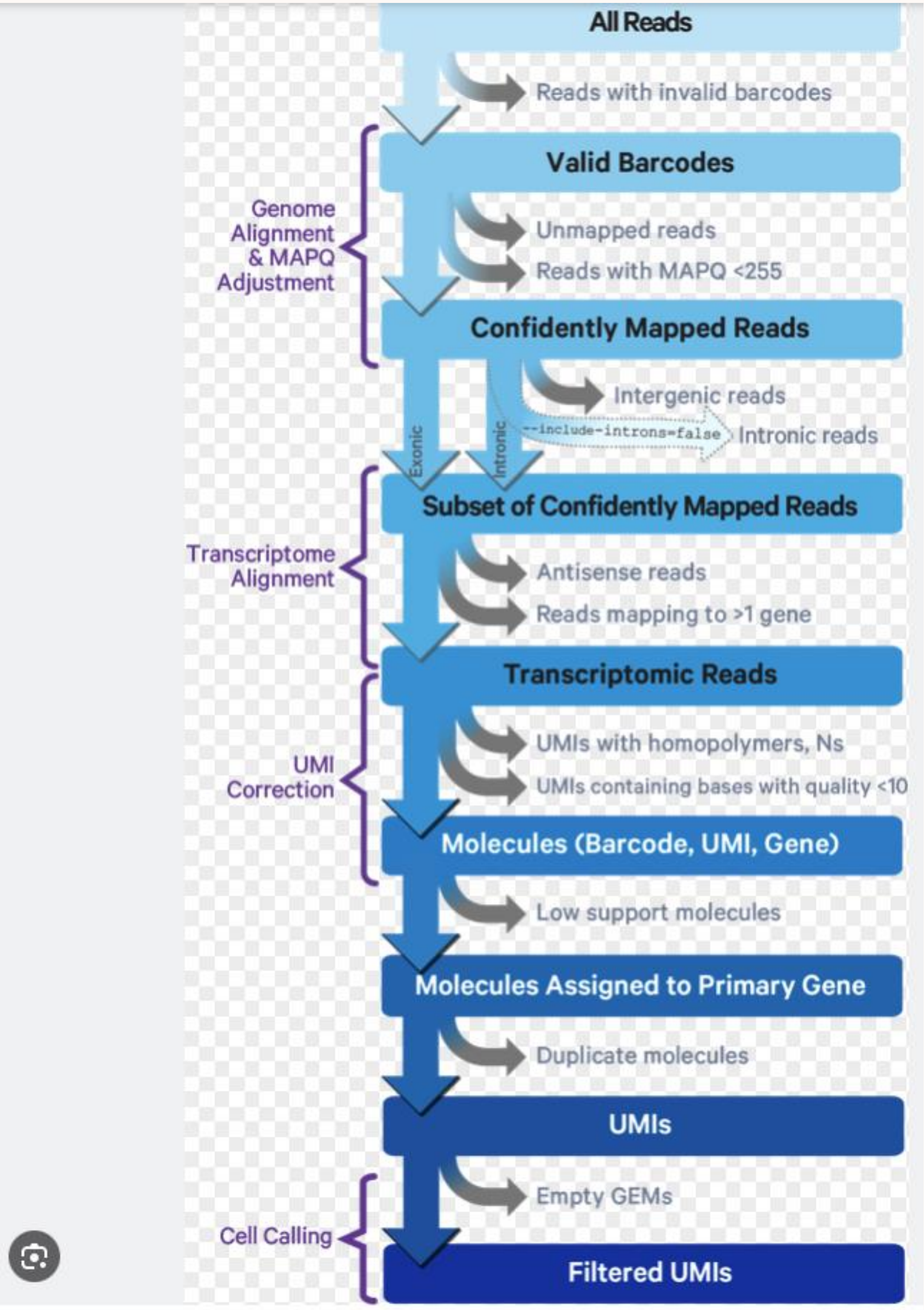
10X Genomics Chromium
Read mapping and counting

Cellranger Pipeline

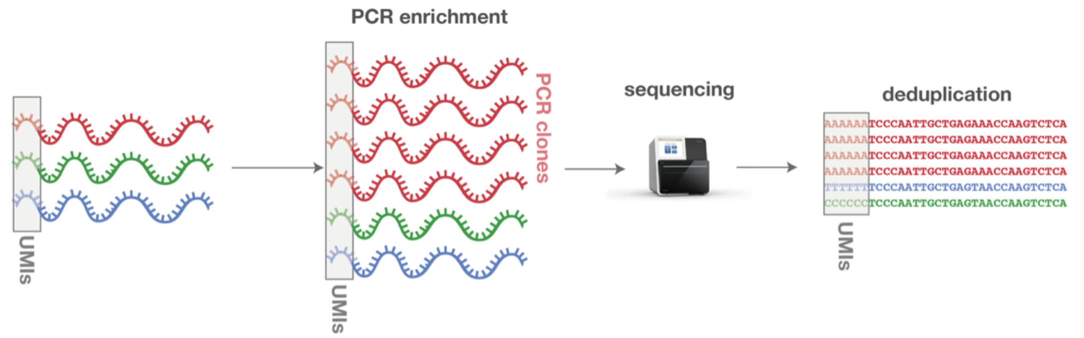
Get
gene x cell
count table



Cell Ranger's Gene Expression Algorithm - Official 10x Genomics Support

UMI: unique molecular identifiers

- ➢ Gene counts are UMI count
- ➢ Remove PCR introduced bias



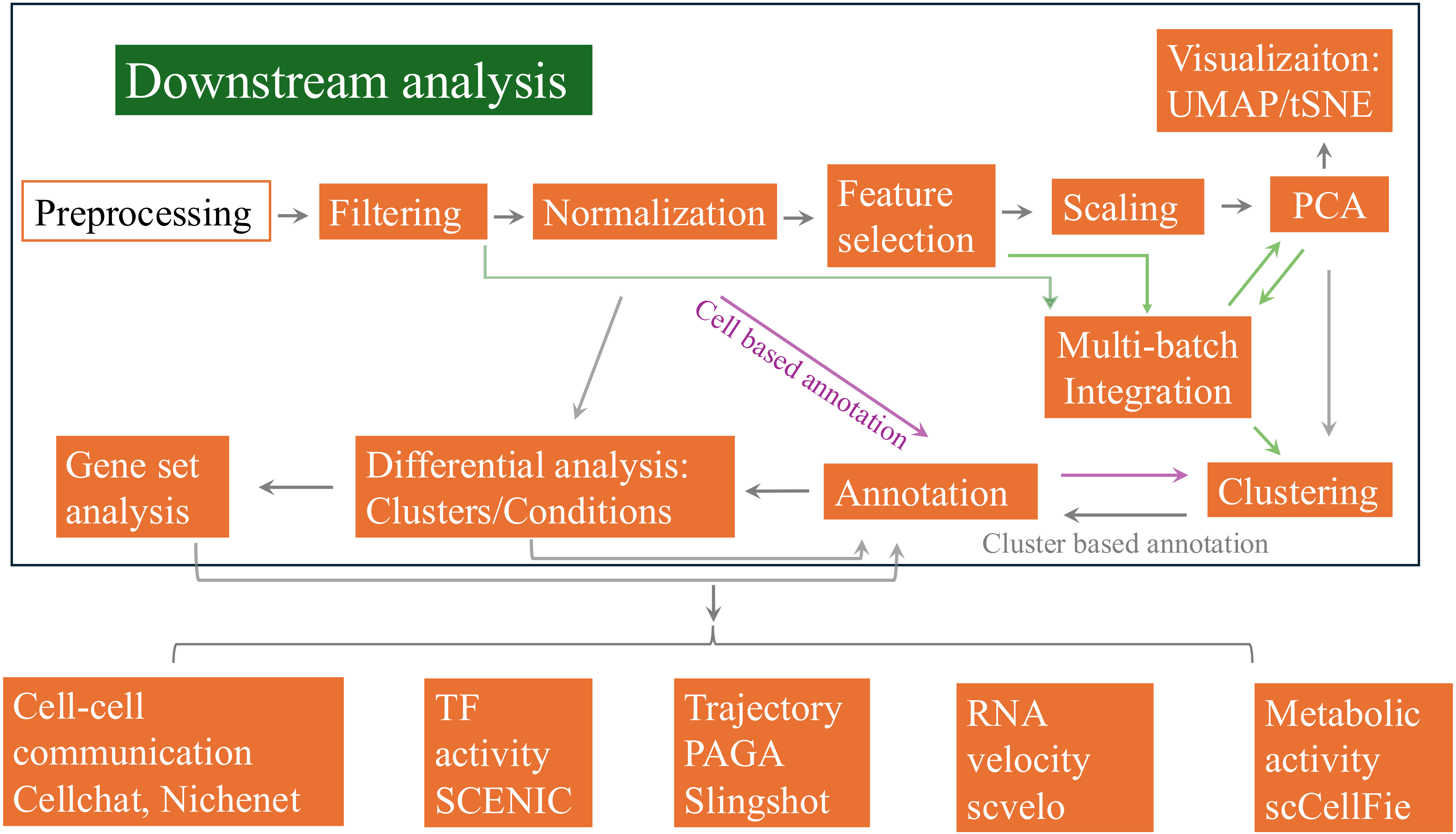
https://www.ecseq.com

Count matrix

	Cell1	Cell2	..
Gene1	0	0	0
Gene2	0	1	0
Gene3	6	0	0
…	0	0	0

Pair discussion:

- **10X Single cell vs common Bulk RNAseq:**
  - How they differ in mRNA quantification in general
    - UMI
    - Cell barcode



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Downstream analysis_popular tools

- **Seurat** (R)
- Scanpy (python)
- Scran (R)

Data variation

Technical

- Cell quality (dying cells, broken cells…)
- Library prep efficiency (RNA capture, sequencing…)
- Batch effects
- …

Biological

- Cell cycle
- Sex, Age
- Transcriptional bursting
- Cell size
- Cell type/state
- …

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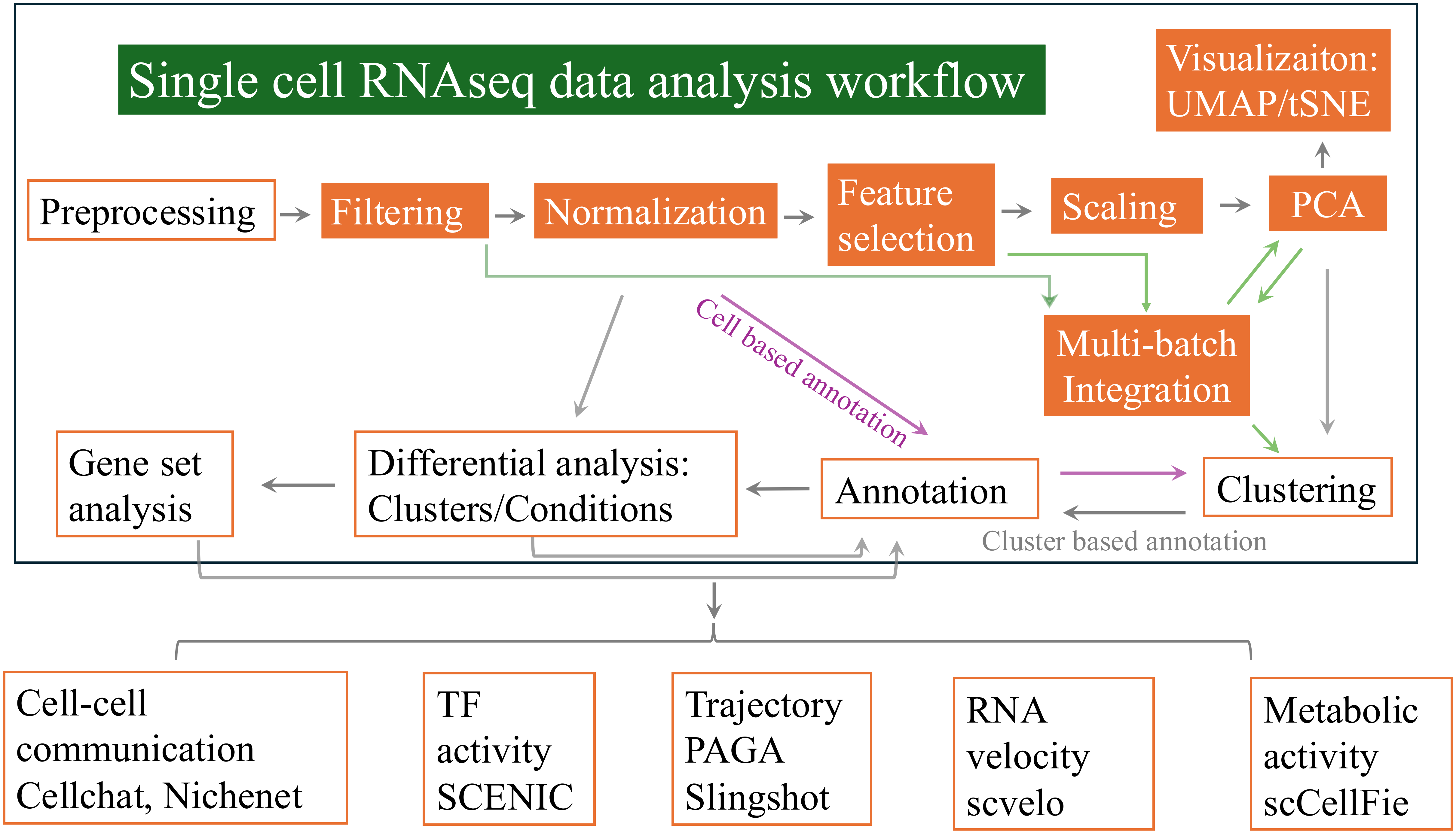
Remove unwanted variation
to look for biological celltypes/states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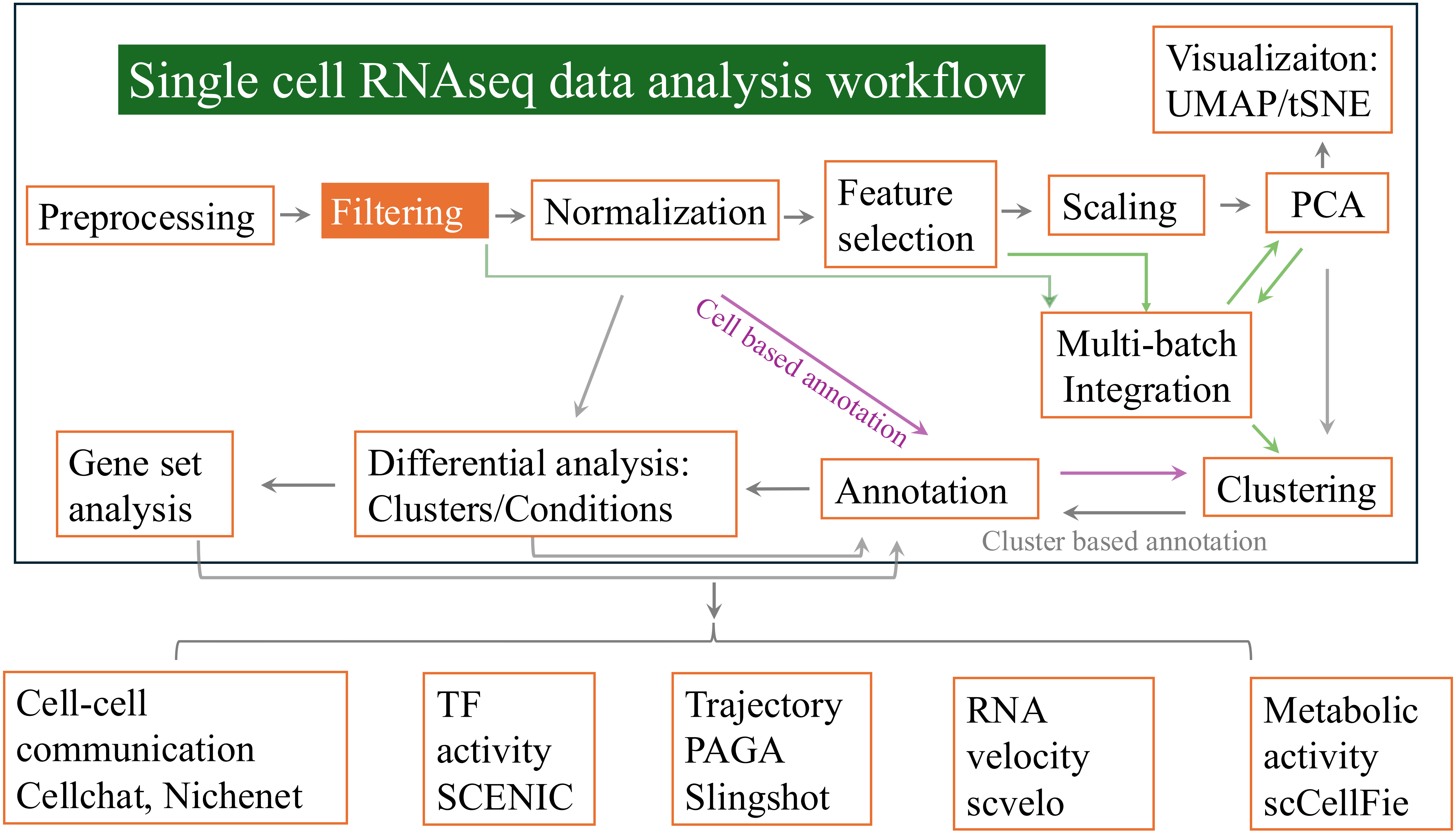
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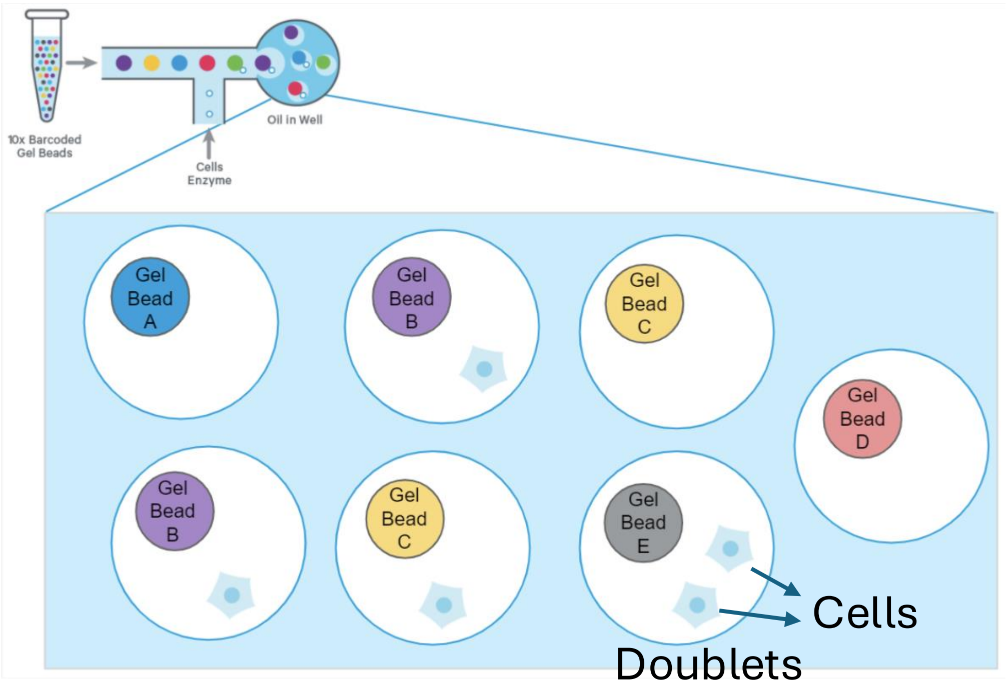
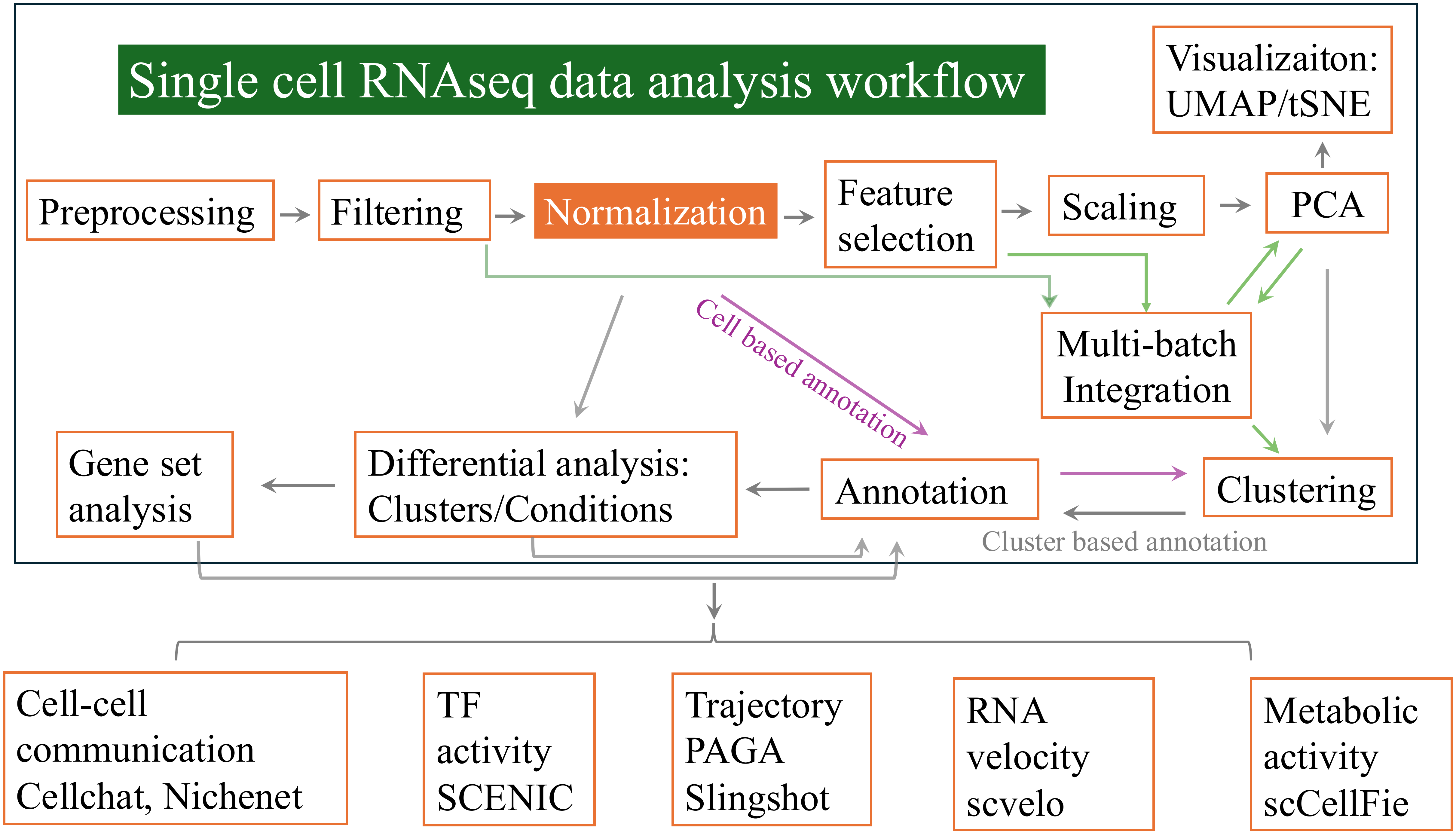
Filtering

- ❖Filter out low-quality genes
  - ➢Lowly captured genes (e.g. <=**3** cells)
- ❖Filter out RNA contamination
  - ➢**Cell-free, ´ambient´ RNA within cell suspension**
- ❖Filter out low-quality cells
  - ➢**Empty droplets** (few genes/UMI)
  - ➢**Low viable cells** (few genes/UMI)
  - ➢**Doublets (too many genes/UMI)**
  - ➢**Broken cells** (leaking, scRNAseq)
    - High mitochondrial gene percentage & few genes/UMI
    - Cytosolic trancripts easier to leak out than mitochondria (big)

Doublet: a 10X droplet capturing 2 cells



https://kb.10xgenomics.com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Normalization

Sampling effect in
- RNA captured by beads
- RNA —> cDNA
- Sequencing of cDNA

Cell cycle
Cell size
Transcriptional bust
Batch effect

➡ Total cell UMI count differ among cells

Hafemeister & Satija 2019 20:296

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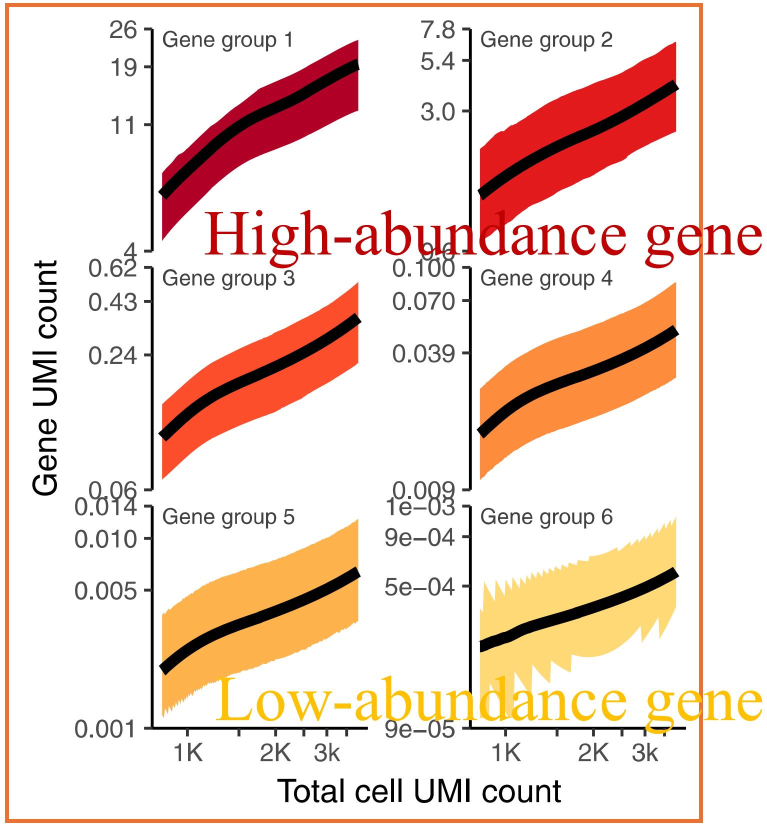

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➢ Gene abundance depends on total cell UMI count **(desired biological differences?)**

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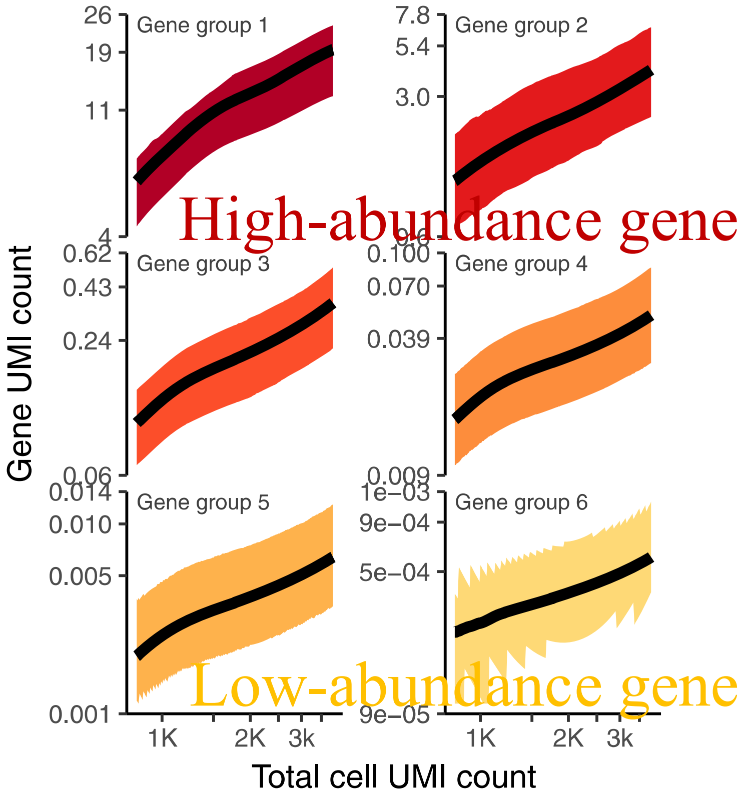


Hafemeister & Satija 2019 20:296

➢Gene abundance depends on total cell UMI count (**desired biological differences?**)

➢ **Normalization**
- (try to) Remove total UMI count differences among cells
- Make gene expression comparable between cells

Normalization methods

- ➢ Seurat: LogNormalize (Scaling)

  * *Assume all cells with the same molecular composition*

LogNormalize() Normalize counts and log-transform
FindVariableFeatures() Select genes with high biological variability
ScaleData() Center & scale data, regress out confounders

RNA:count slot: raw count

Seurat::NormalizeData()

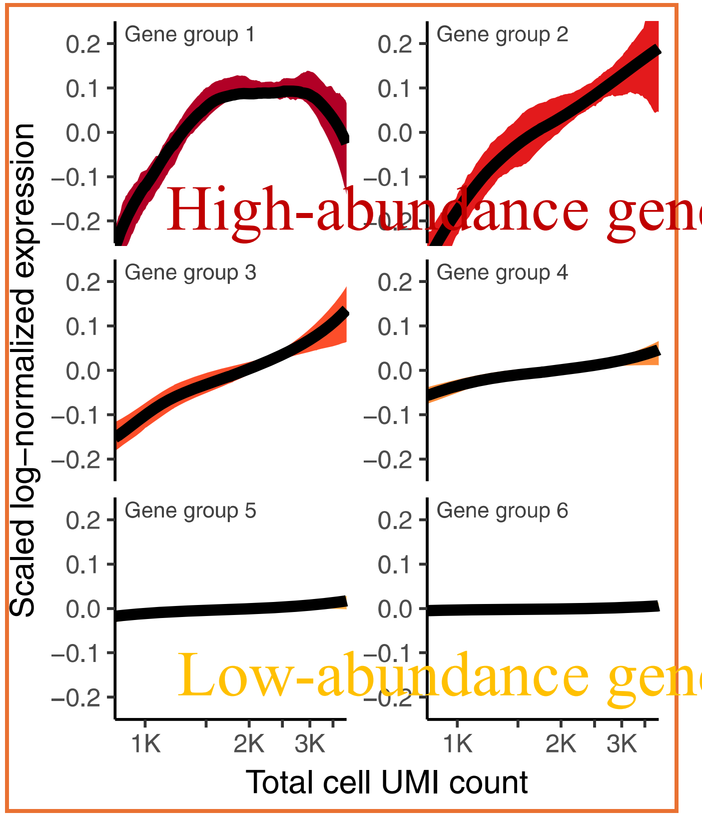
Each cell's counts are:

1. **Divided by the total counts** for that cell (to no
2. **Multiplied by a scale factor** (default = 10,000)
3. **Log-transformed** (using natural log).

$$x'_{ij} = \log\left(\frac{x_{ij}}{\text{total}_j} \times 10{,}000 + 1\right)$$

- x_{ij}: raw count of gene *i* in cell *j*
- total_j: total counts in cell *j*

RNA:data slot:
* Normalized count
* Near normal distribution
* All cells the same total UMIs



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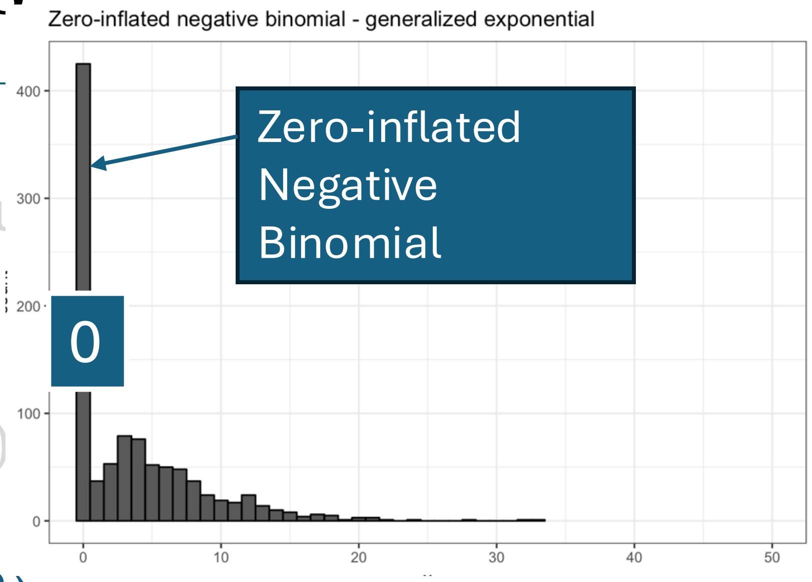
Normalization methods

- Seurat: LogNormalize (Scaling)
  * Assume all cells with the same molecular composition
- High-count filtering CPM
- BASiCs (Spike-in based)
- Scran (Lun et al. 2016) (Pooling-based)
- **Downsampling**

Normalization methods



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- Downsampling
- ZINB-Wave (non-UMI, e.g. SmartSeq2)
- **Seurat: SCTransform v2 (with seq depth, UMI follow NB)**
  - **Regularized Negative Binomial Regression**

- Model parameters may differs between datasets: normalize separately
- **All 3 steps in one function (sctransform()):**

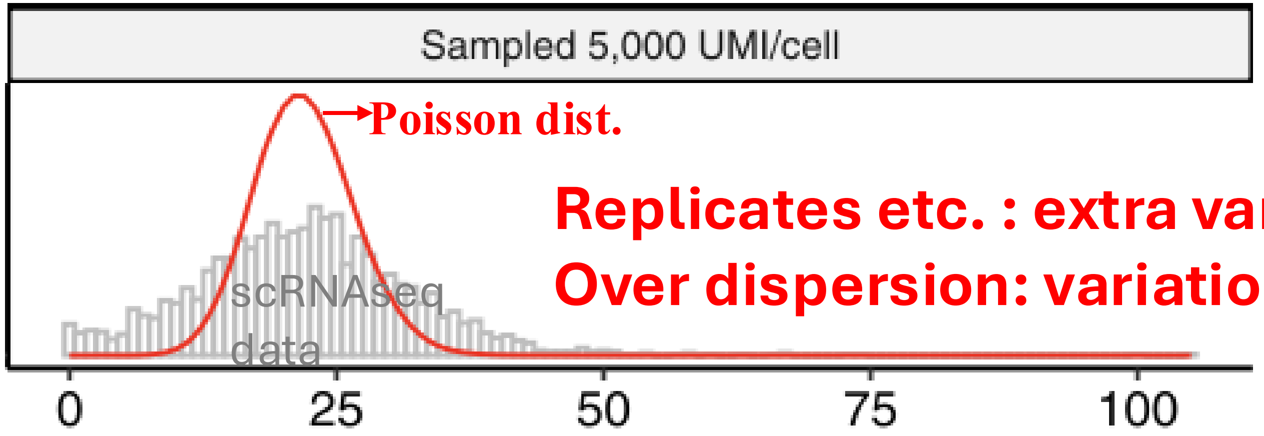
  LogNormalize() Normalize counts and log-transform

  FindVariableFeatures() Select genes with high biological variability

  ScaleData() Center & scale data, regress out confounders

Single cell RNAseq data modeling:

- ❖ **Count: poisson distribution (variation=mean)**



Replicates etc. : extra variation to count data

Over dispersion: variation>mean

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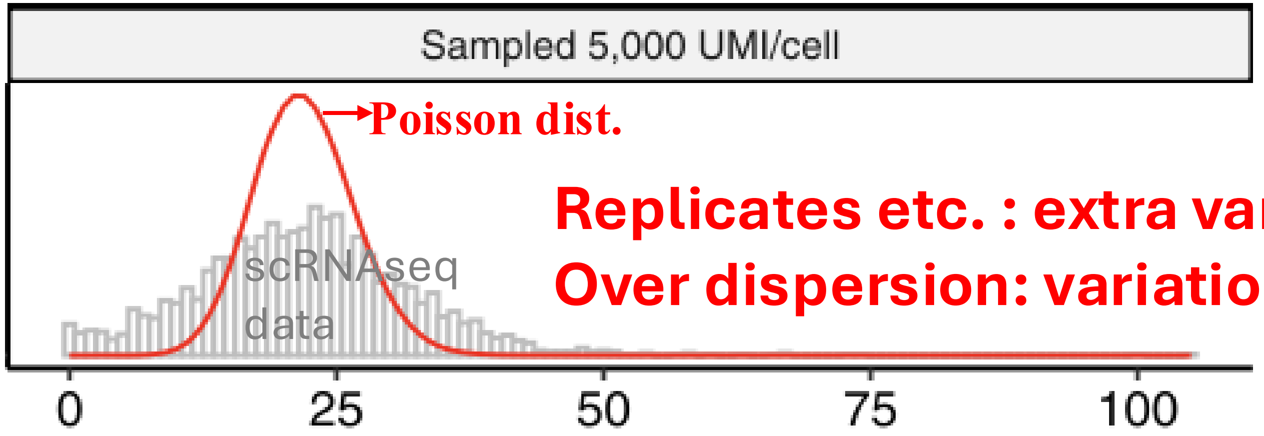

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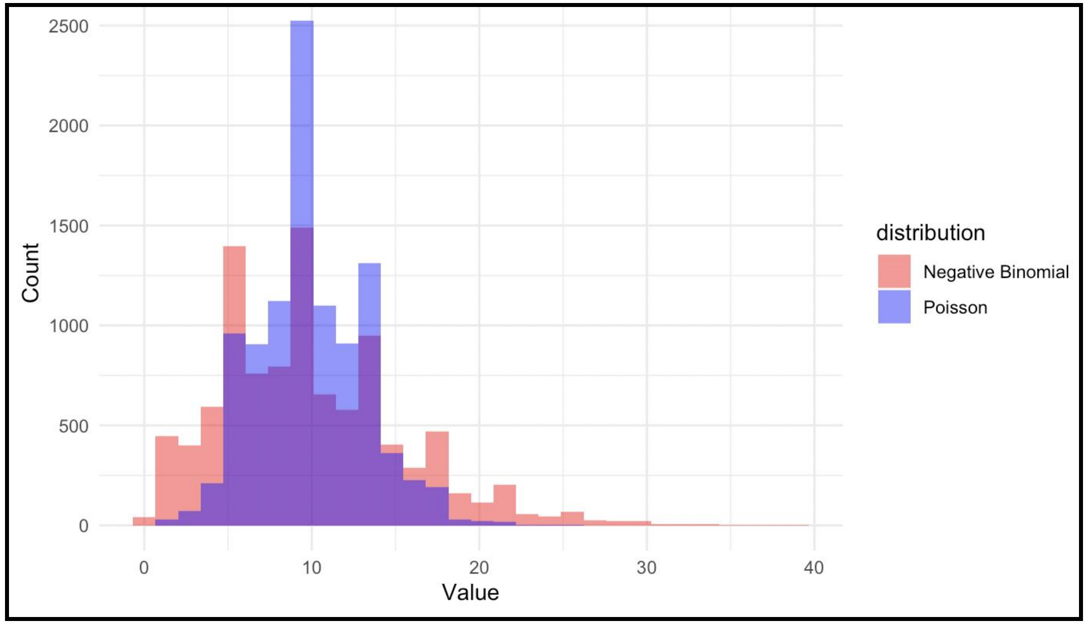

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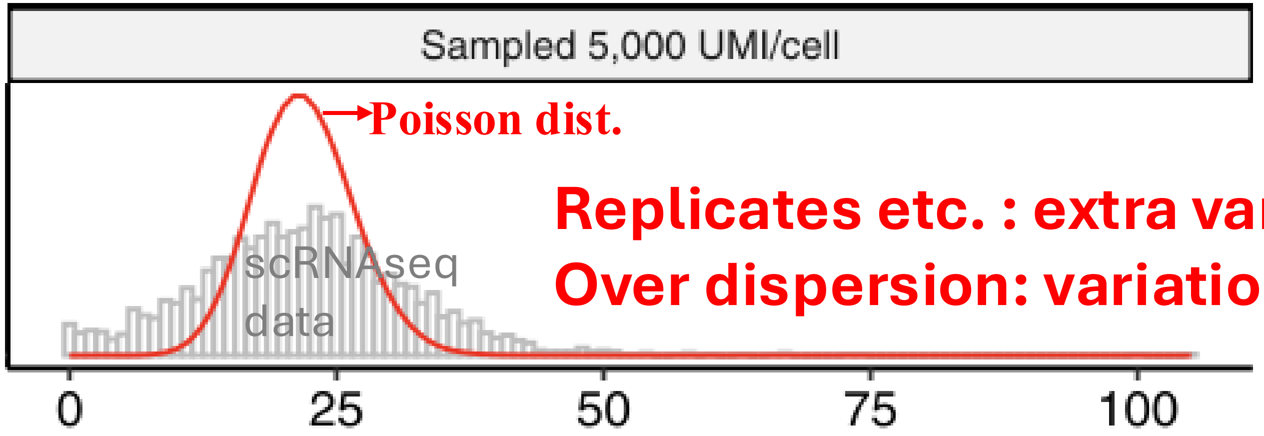

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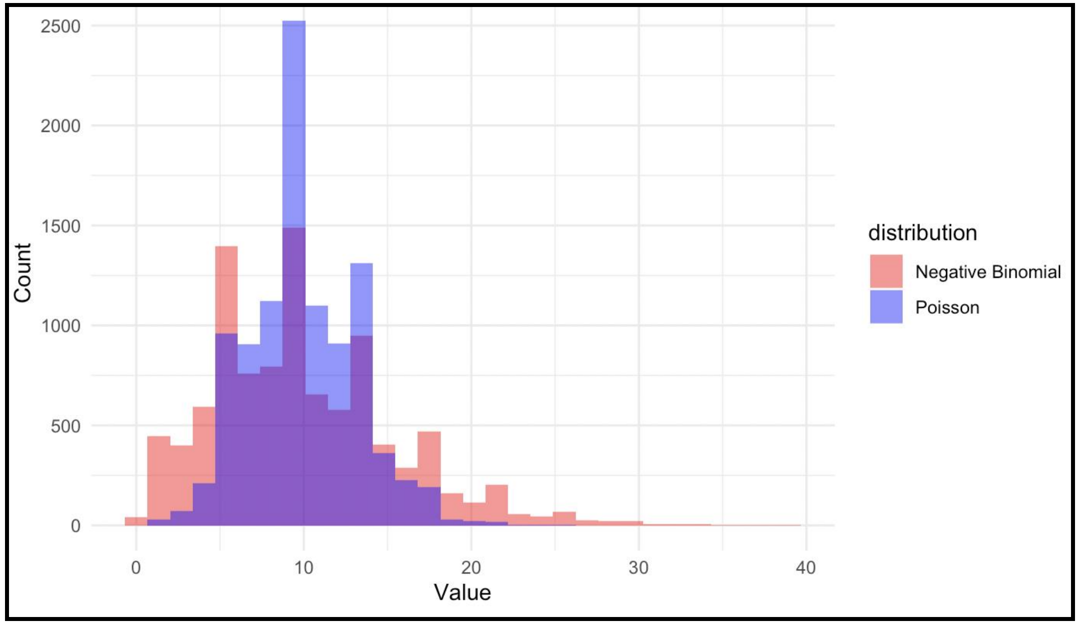

NB GLM:

Total cell UMI

$$\log(\mathbb{E}(x_i)) = \beta_0 + \boxed{\beta_1 \log_{10} m_i}$$

$y_i = NB(E(x_i), \theta)$

Dispersion parameter

Remove total cell UMI dependence

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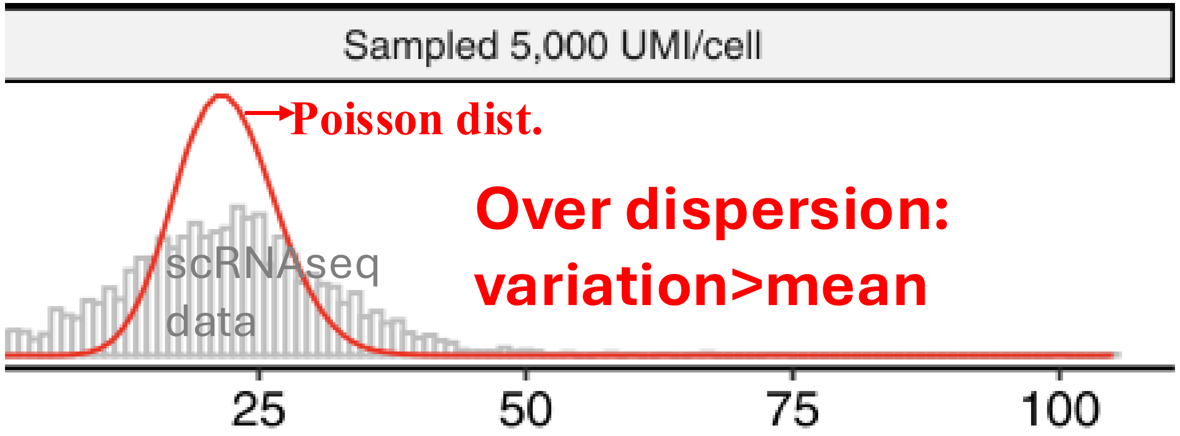

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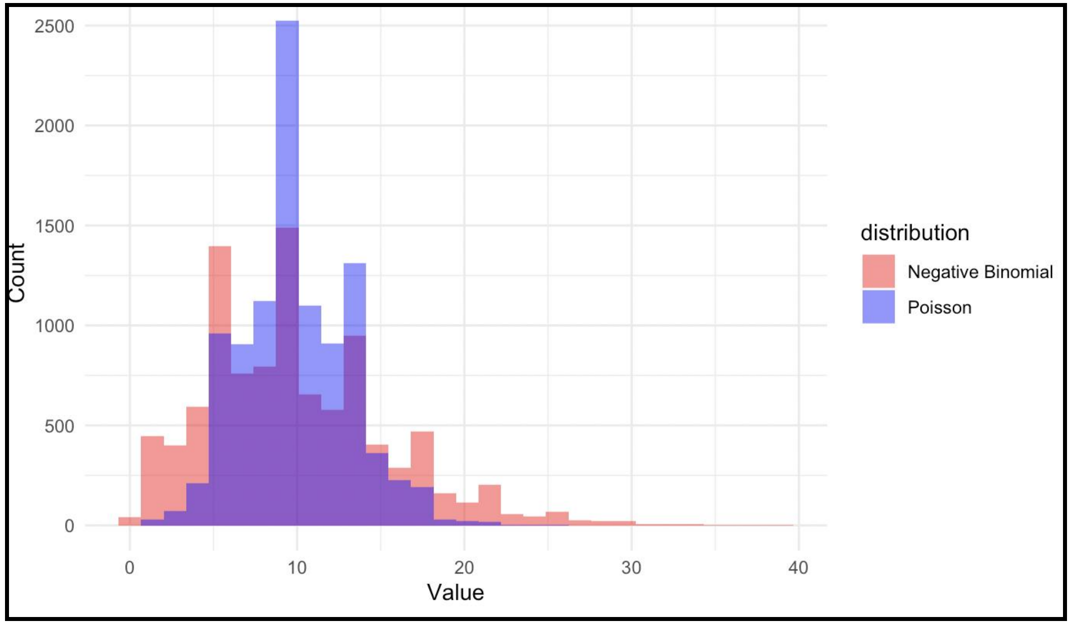

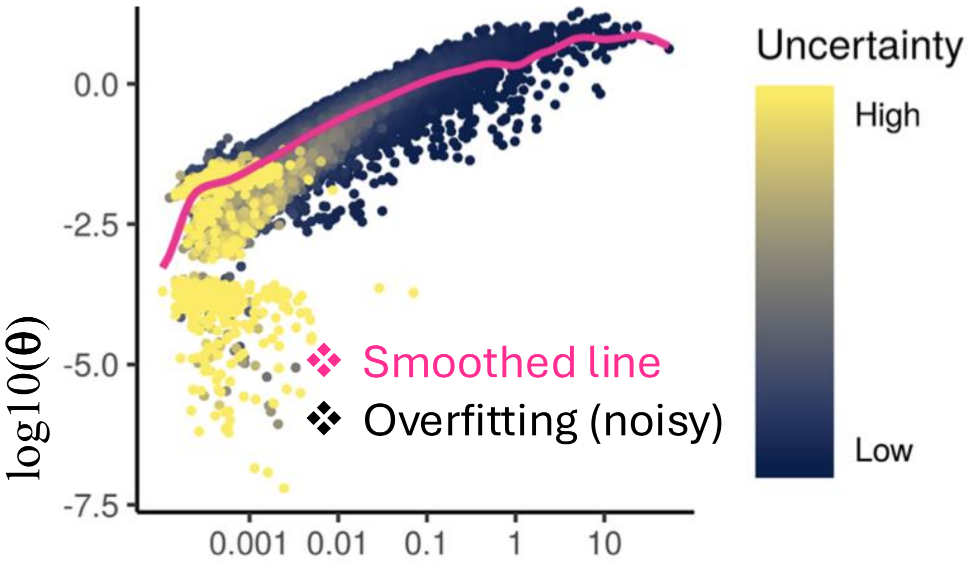

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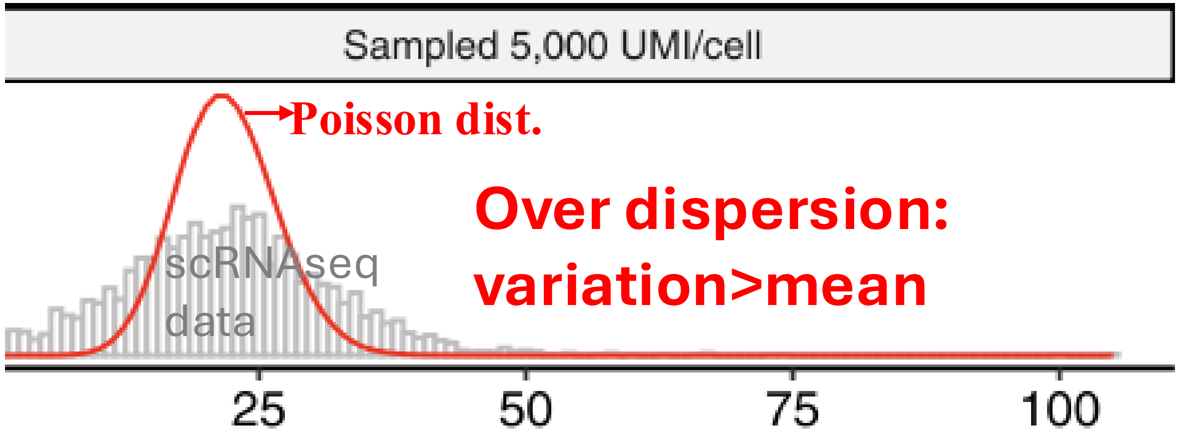

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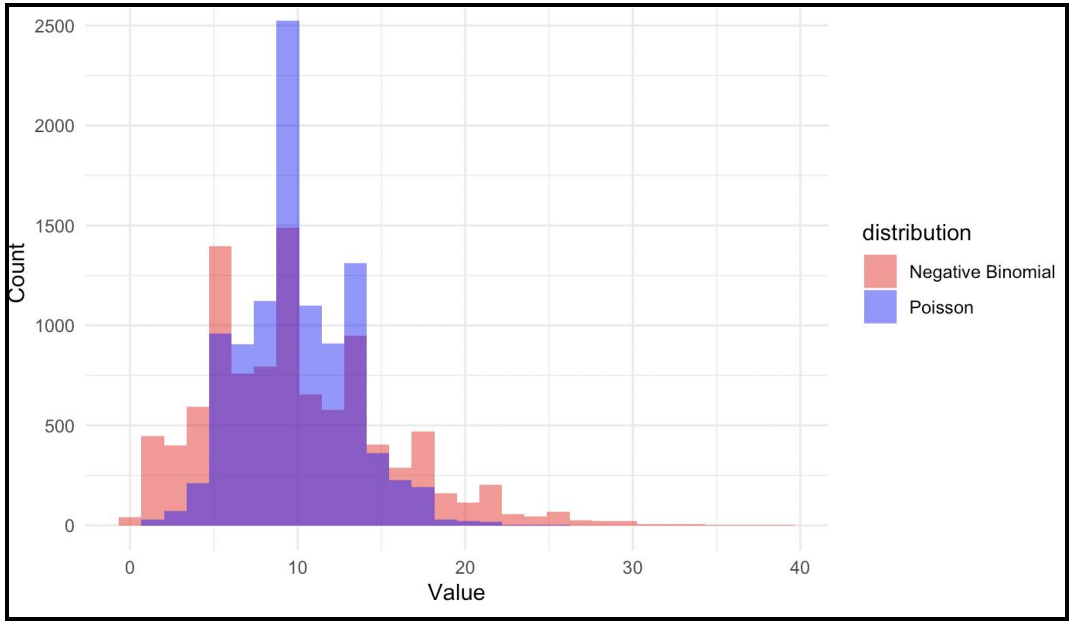
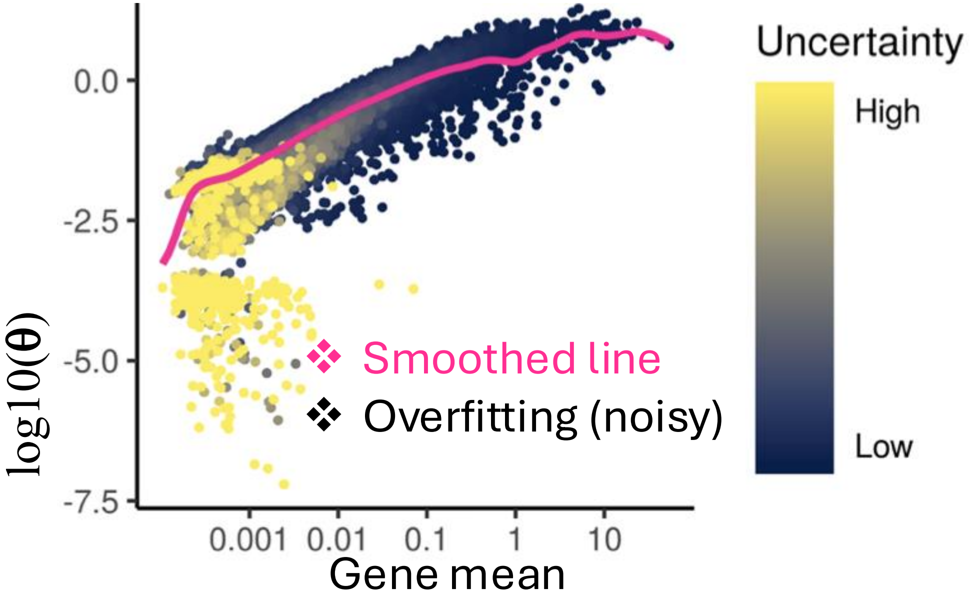


Model parameter Regularization

* Constrain towards smoothed line
* Increase estimate robustness

Exclude lowly expr. (Poisson) Genes (v2)



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$$\log(\mathbb{E}(x_i)) = \beta_0 + \boxed{\beta_1 \log_{10} m_i} + \text{Covariates}$$

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Pearson residual:

$$r_{ij} = \frac{y_{ij} - \hat{\mu}_{ij}}{\underbrace{\sqrt{\hat{\mu}_{ij} + \hat{\mu}_{ij}^2/\theta_i}}_{\text{NB SD}}}$$

SCT:scale.data slot:

* No overdispersion
* Nearly normal
* mean = 0, variation stabilized
* total UMIs (+covariates) effect removed
* Ready for PCA

SCT:count slot:

* Neg. binomial distribution
* overdispersion (variance > mean)
* Variance increase with mean
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$$y_{ij}^{\text{corrected}} = r_{ij} * \text{NB SD} + \exp(\beta_0)$$

SCT:data slot:

* log transformed
* near normal
* total UMIs (+covariates) effect removed

Pair discussion:

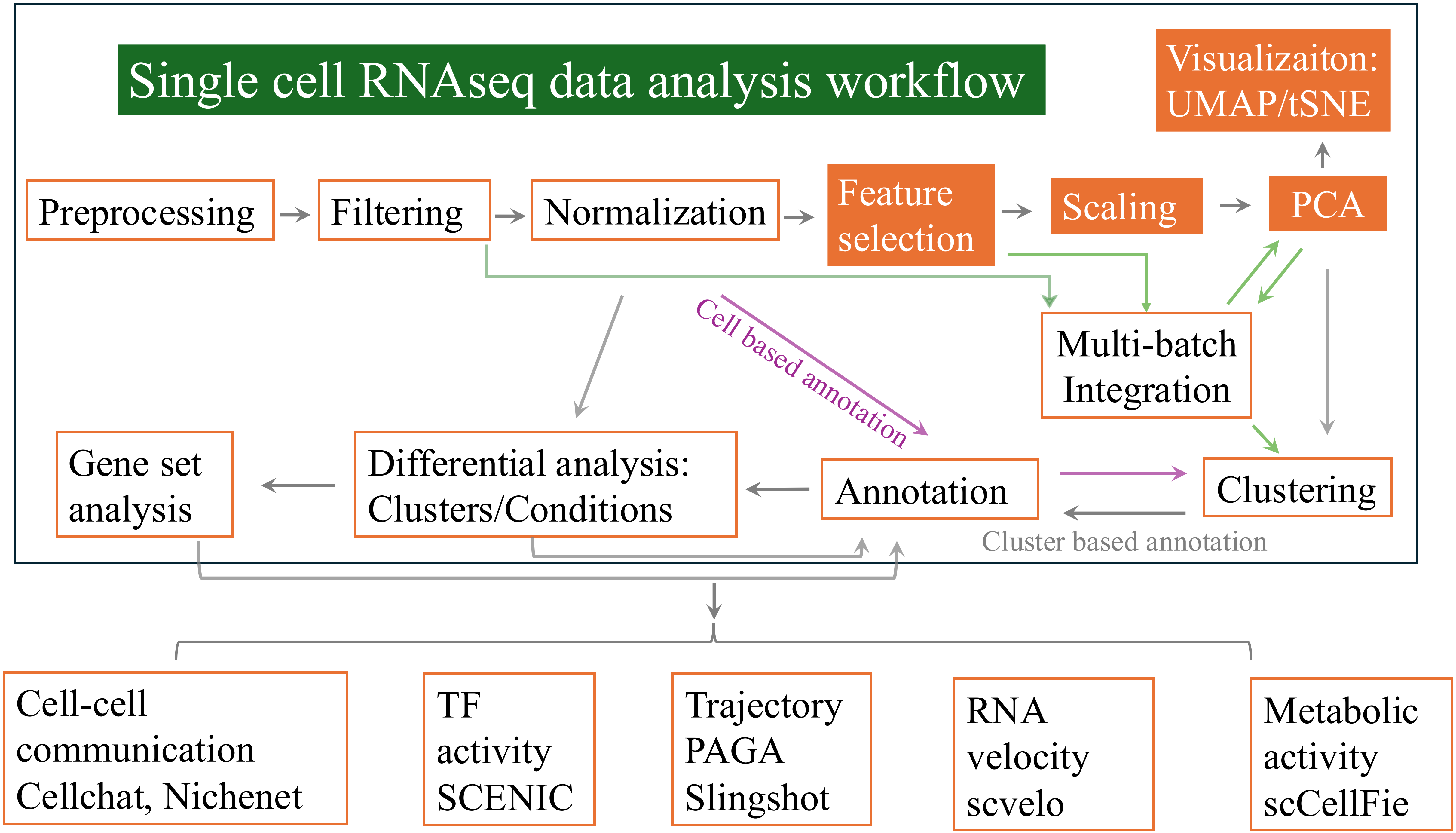
- Why do we do normalization?
- What kind of data do the 3 data slots have in the Seurat SCT assay:
  - count slot:
  - data slot:
  - scale.data slot:

Pair discussion:

- Why do we do normalization?
  - Remove the dependency between gene abundance and total cell UMI count
  - Make genes comparable among cells

- What kind of data do the 3 data slots have in the Seurat SCT assay:
  - count slot: corrected count, with total cell UMI count (+covariate) effect remove, NB
  - data slot: log transformed corrected count
  - scale.data slot: standardized residuals, scaled data

Dimensional reduction



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High dimensional curse

Single cell data: high dimensional data (P genes = P dimesions, thousands of genes)

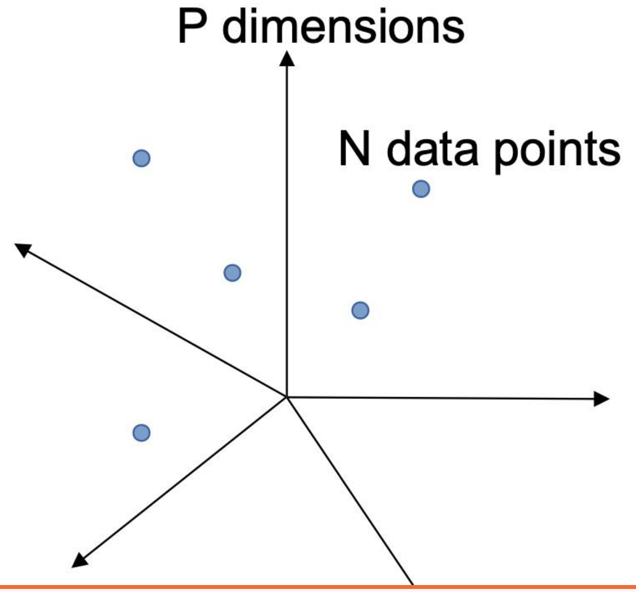
Statistical observations: e.g. samples, cells etc.

N

Dimensions: P

Features: genes, proteins, microbes, metabolites etc.

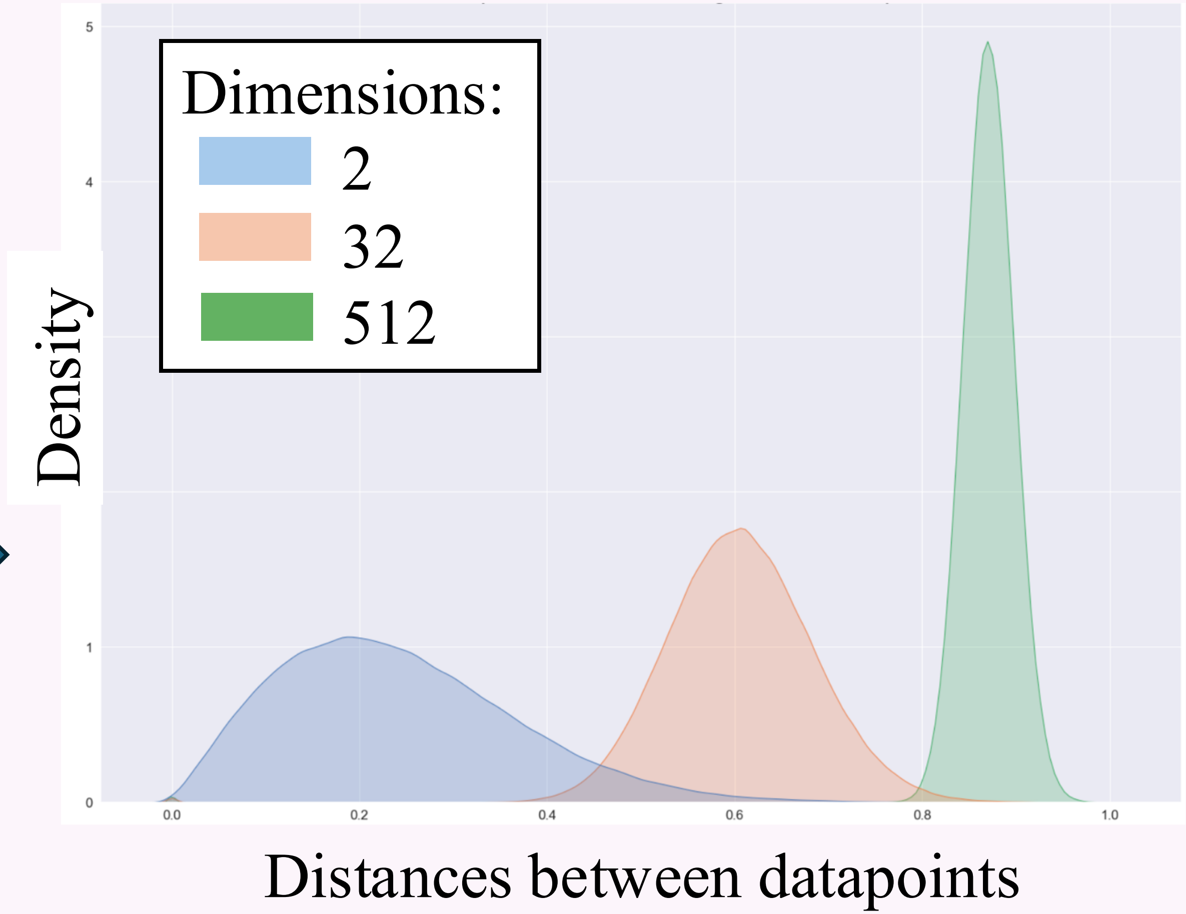
$$\begin{pmatrix} 0 & 3 & 1 & 0 & 2 & 3 & 8 & 1 & 1 & 3 \\ 1 & 1 & 0 & 0 & 7 & 1 & 2 & 2 & 3 & 3 \\ 1 & 2 & 2 & 0 & 0 & 6 & 7 & 1 & 2 & 2 \\ 1 & 2 & 3 & 10 & 0 & 4 & 6 & 1 & 0 & 5 \\ 3 & 2 & 2 & 1 & 4 & 3 & 2 & 1 & 6 & 0 \\ 7 & 4 & 4 & 5 & 3 & 9 & 6 & 1 & 6 & 1 \\ 7 & 1 & 1 & 5 & 2 & 8 & 9 & 1 & 3 & 6 \\ 5 & 0 & 1 & 6 & 2 & 0 & 0 & 0 & 1 & 5 \\ 1 & 6 & 3 & 3 & 4 & 6 & 2 & 0 & 1 & 1 \\ 1 & 2 & 2 & 4 & 1 & 1 & 3 & 0 & 8 & 2 \end{pmatrix}$$



High dimensional curse

Curses:

- Data sparsity (missing values, noisy)
- Multicollinearity (correlation between genes)
- Multiple testing
- Overfitting (multivariate analysis)
- Data points very far from each other (**hard to form clusters/celltypes! Visualization**)



High dimensional curse

- **Solutions:**
  - Select (e.g. 2k) highly variable genes/HVGs (less genes/dimensions)
  - **Dimensional reduction**

    (**Compress** 2k HVGs into **fewer** dimensions/`MegaGenes`)

    * (Scaling)

    - PCA
    - tSNE
    - UMAP

Select highly variable genes

Assumptions:

- Features (i.e. genes) with low variance: **noises**
- Features with high variance: biological factors

Variation stabilization (Variance>mean)!

- Seurat:sctransform()
    - Based on the residual variance

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SCT:data slot:
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* near normal
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Scaling

- Why do scaling?
  - PCA/tSNE/UMAP: multivariate analysis
  - Require features to be on similar scales

- Seurat sctransform()
  - Pearson residuals (variance stabilized residuals)

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Dimensional reduction

- **PCA**:
  - Linear transf.(accurately represent distances between cells)
  - Cannot efficiently pack differences from high dim. into the top PCs (principal components)-formed space
  - Capture global structure
- **tSNE**
  - Preserve local structure
  - Perplexity, low: finer structue
- **UMAP**:
  - Preserve both local & global structure
  - n_neighbors, min_dist, low: finer structure
- **tSNE&UMAP**
  - Non-linear (discard inf. distort distance)
  - NOT accurately represent distances between cells

Quantification

Only for visualization

Moon et al. 2019 Nature Biotechnology

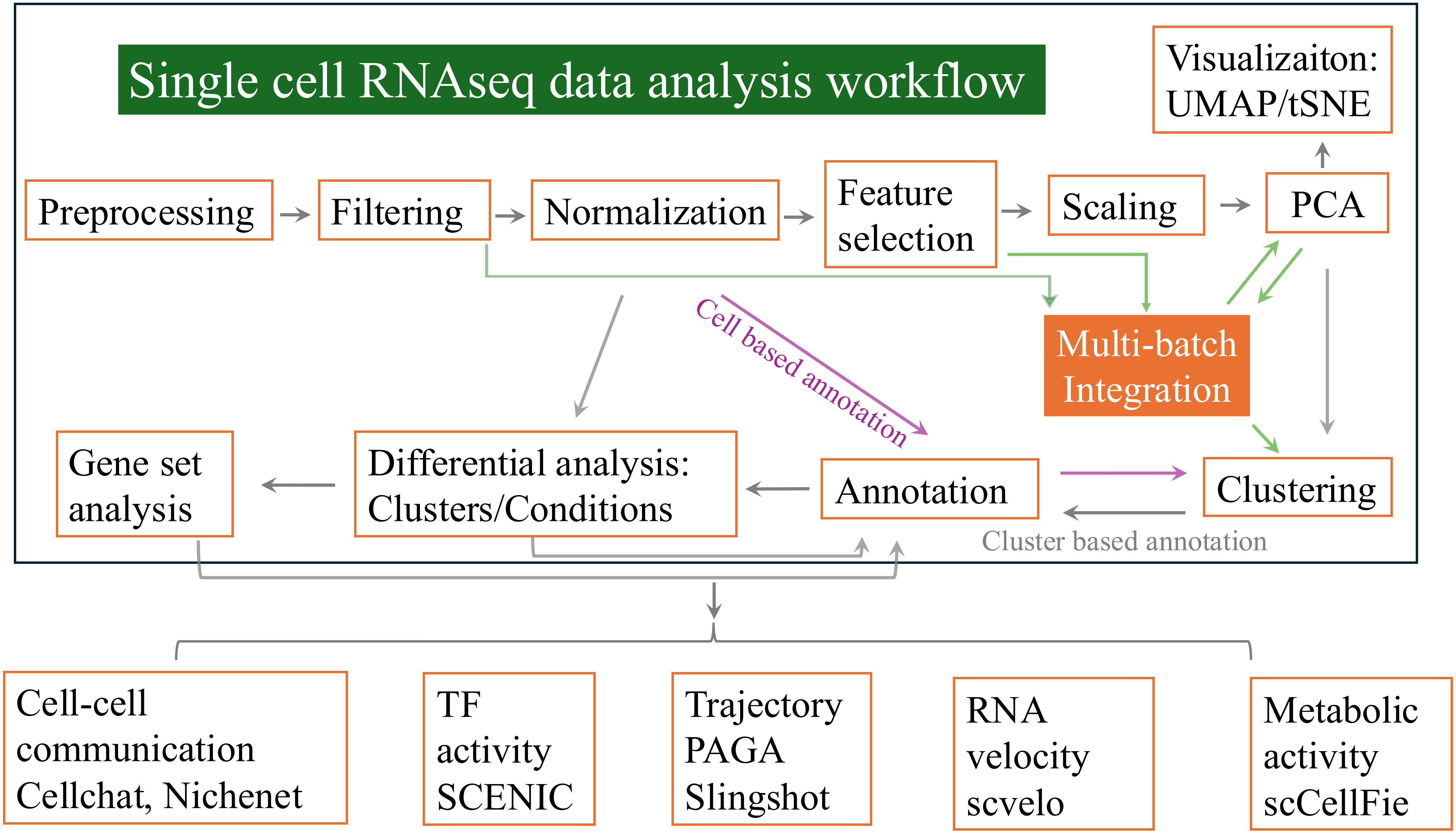
Pair discussion:

- When select highly variable genes, why need to stabilize variation first?

Pair discussion:

- When select highly variable genes, why need to stabilize variation?
  - scRNAseq data: NB
  - Variance increase with mean

What if there are multiple samples from different batches?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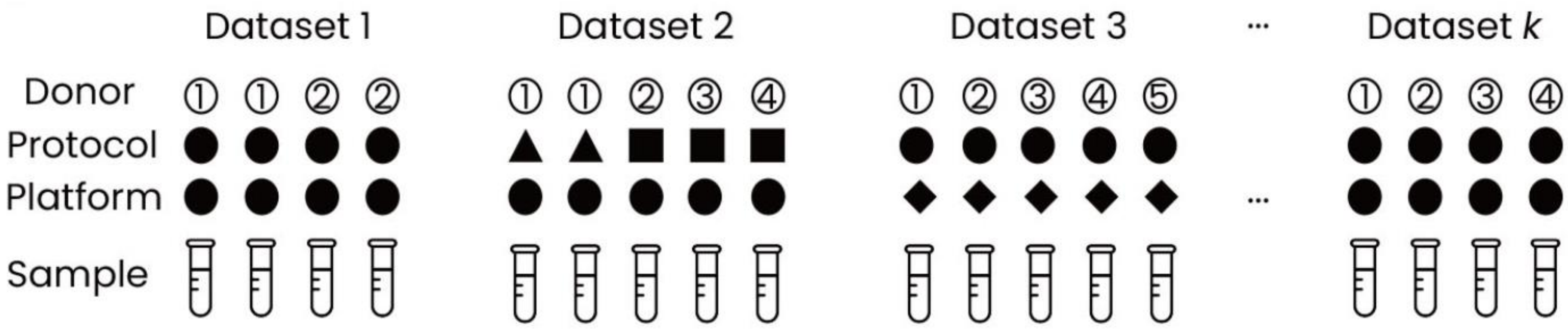
PCA
Cell based annotation
Multi-batch
Integration
Gene set
analysis
Differential analysis:
Clusters/Conditions
Annotation
Clustering
Cluster based annotation
Cell-cell
communication
Cellchat, Nichenet
TF
activity
SCENIC
Trajectory
PAGA
Slingshot
RNA
velocity
scvelo
Metabolic
activity
scCellFie

Integration: remove batch effect

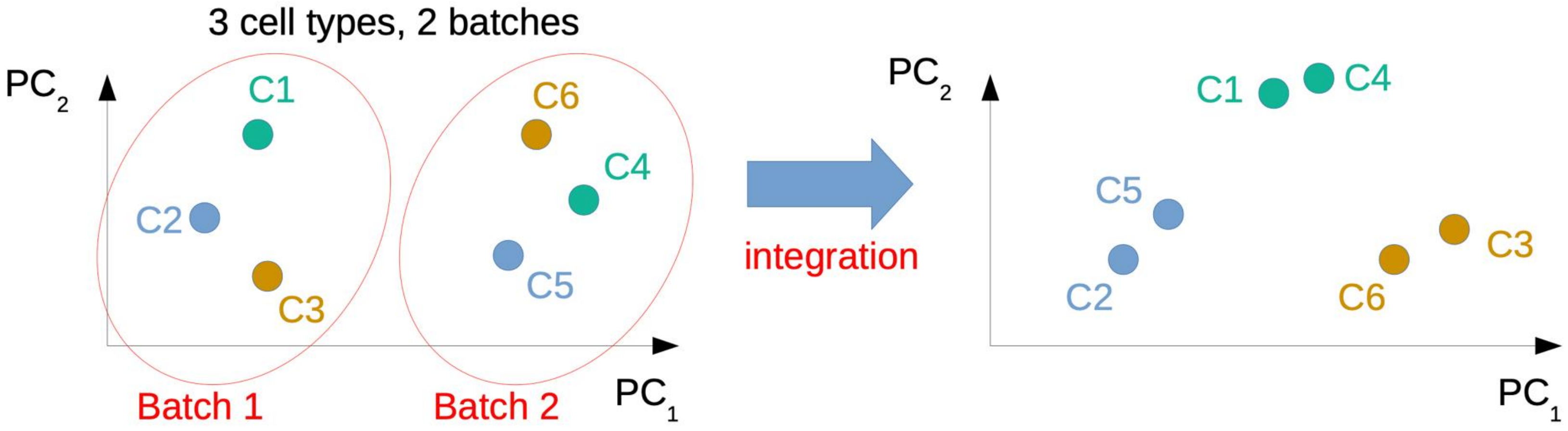
Batches:

- Sample characteristics (donors, tissues, species) …
- Experimental protocols, laboratories, sample handling, reagents or media and/or sampling time …
- Sequencing platforms/lanes/flow cells …



Ryu et al. Mol. Cells 2023; 46(2): 106-119

Integration: remove batch effect

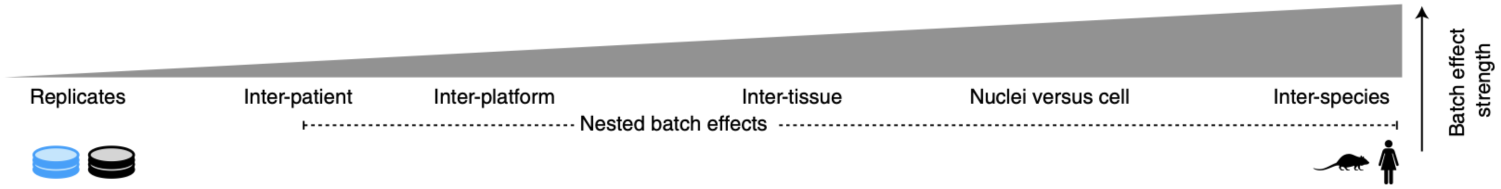


a

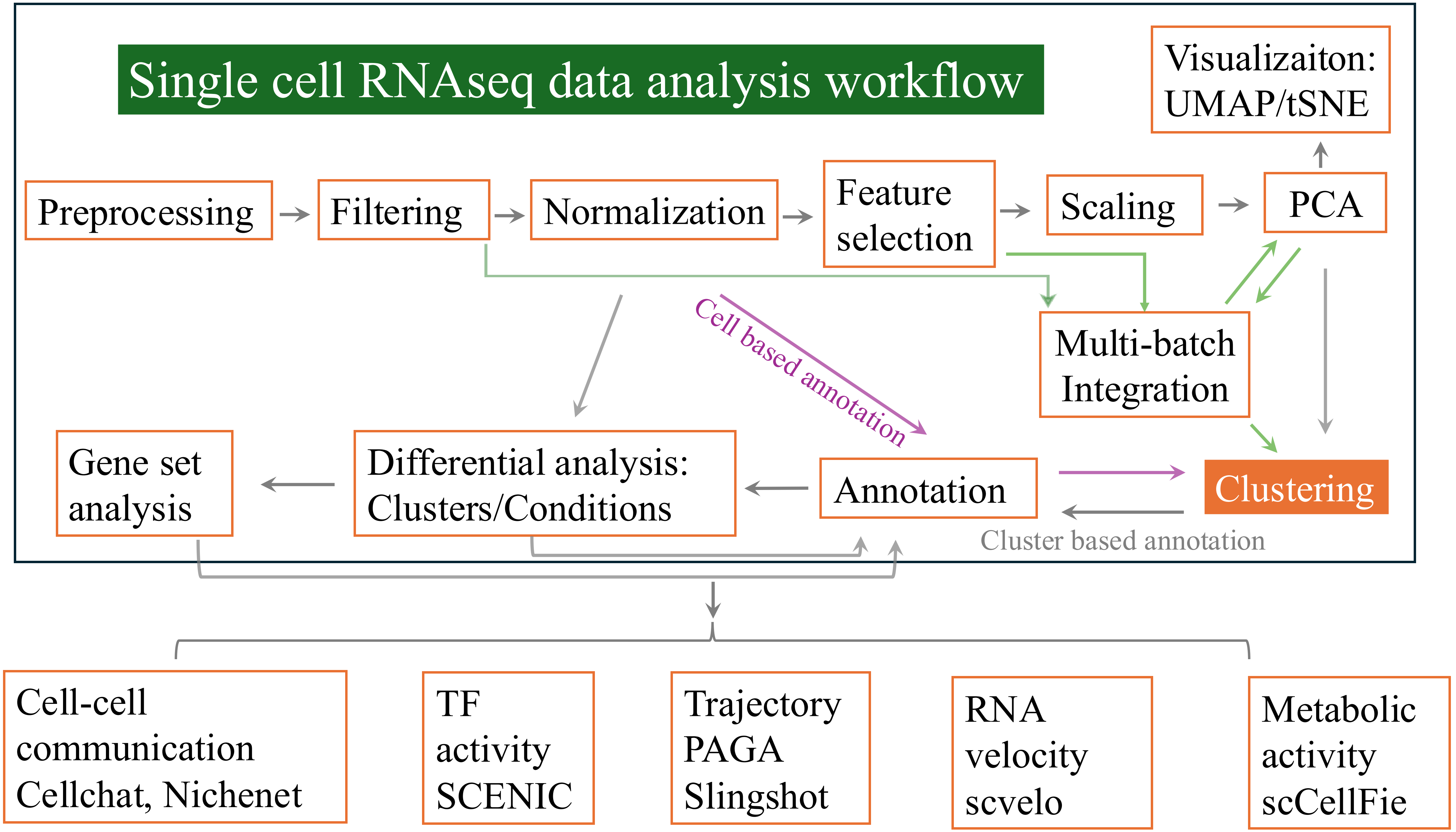
	Considerations	scANVI	Scanorama embed	scVI	FastMNN embed	scGen	Harmony	FastMNN gene	Seurat v3 RPCA	BBKNN	Scanorama gene	ComBat	MNN	Seurat v3 CCA	trVAE	Conos	DESC	LIGER	SAUCIE embed	SAUCIE gene
Input	Programming language	Python	Python	Python	R	Python	R	R	R	Python	Python	Python/R	Python/R	R	Python	R	Python	R	Python	Python
	Method runs without additional information	✗				✗														
Scib results	Consistent top performer	✓	✓	✓		✓														
	Top method on small/ simple tasks		✓		✓	✓	✓													
	Top method on large/ complex tasks	✓	✓	✓		✓														
	Top method on ATAC data	–		–			✓											✓		
Task details	Integrates strong batch effects	✓	–	–		✓			–	–				–						
	Top method for recovery cell states or modules	✓	✓								✓	✓	✓							
	Confounding of bio and batch variance	✓	–			✓														
	Top method for trajectories	–	✓	–	✓	✓														
	Method deals with varying compositions											✗								
Speed	Fast method for quick results									✓		✓								
	Scales well to large datasets on CPU	✓	–	✓						✓	–								✓	✓
	Method has GPU support	✓		✓		✓									✓		✓		✓	✓
	Scales well to feature spaces beyond genes														✓	✓				
Output	Method shows corrected expression					✓		✓	✓		✓	✓	✓	✓						✓
	Method gives relative cell embeddings									✗						✗				

✓ Fulfills the criterion
– Partial fulfillment of criterion
✗ Does not fulfill criterion
Python
R

b



Lueken et al. 2022 Nature Methods



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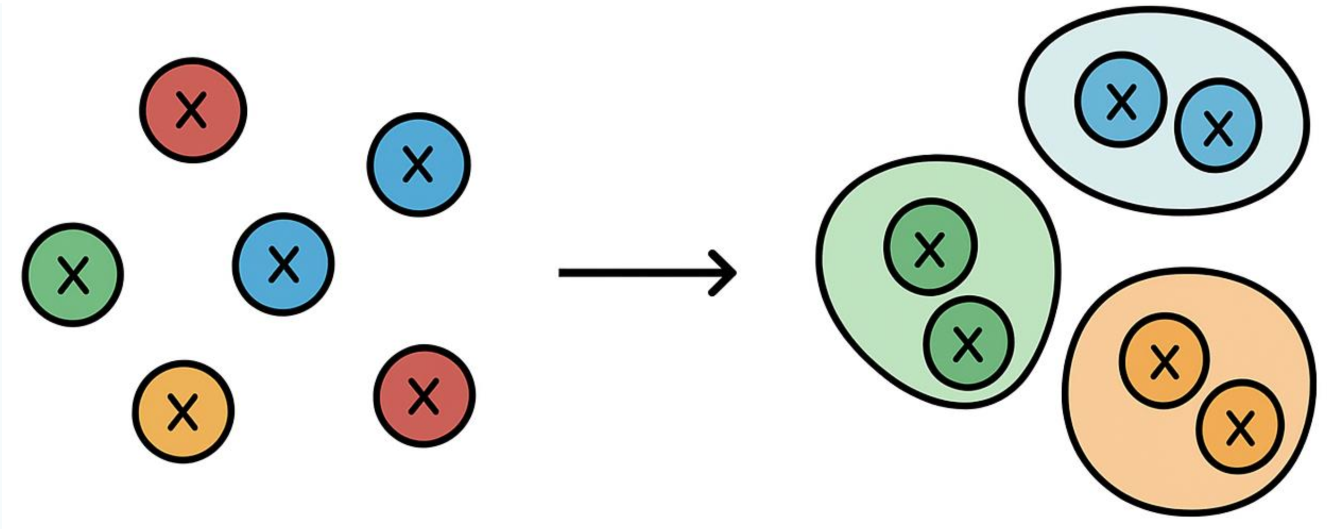
Graph-based

Clustering

Analyze ind. cells?

Group cells to clusters!

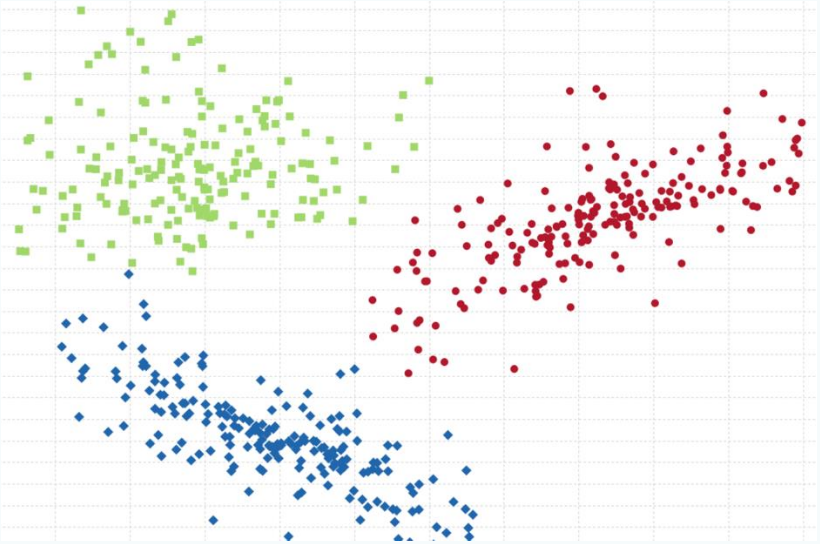


- Noise, variability
- Biological context lost

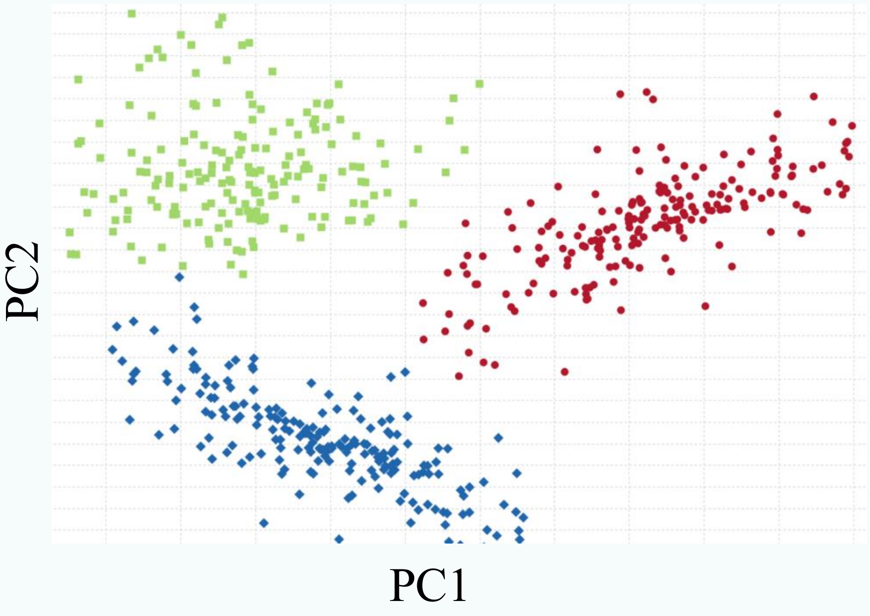
- Reduce noise, robust insights
- Biologically meaningful patterns

Cell clusters/groups $\approx$ Celltypes/cellular states

PCA space: on top n PCs

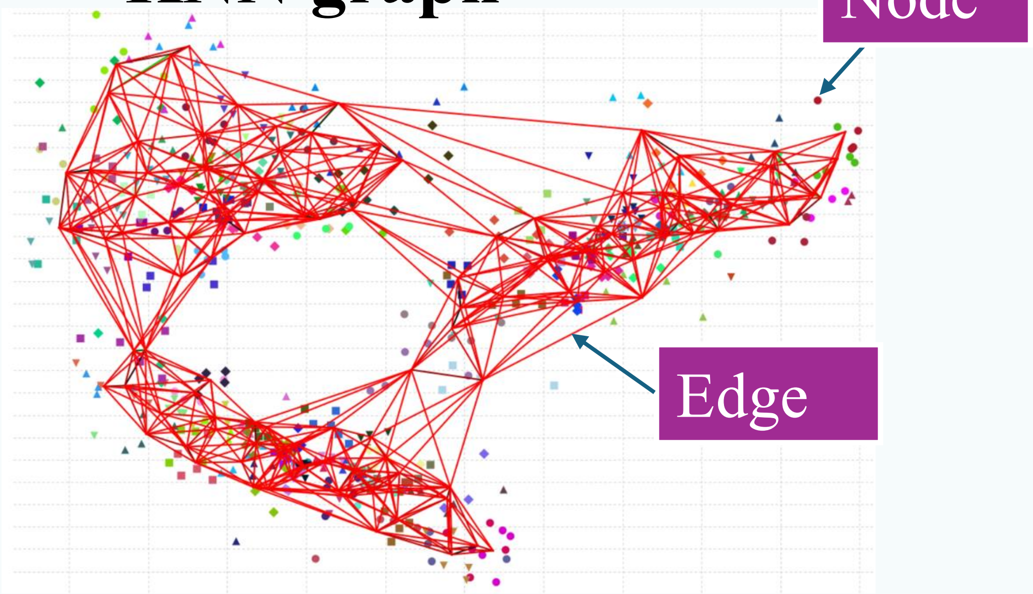


PCA space: on top n PCs

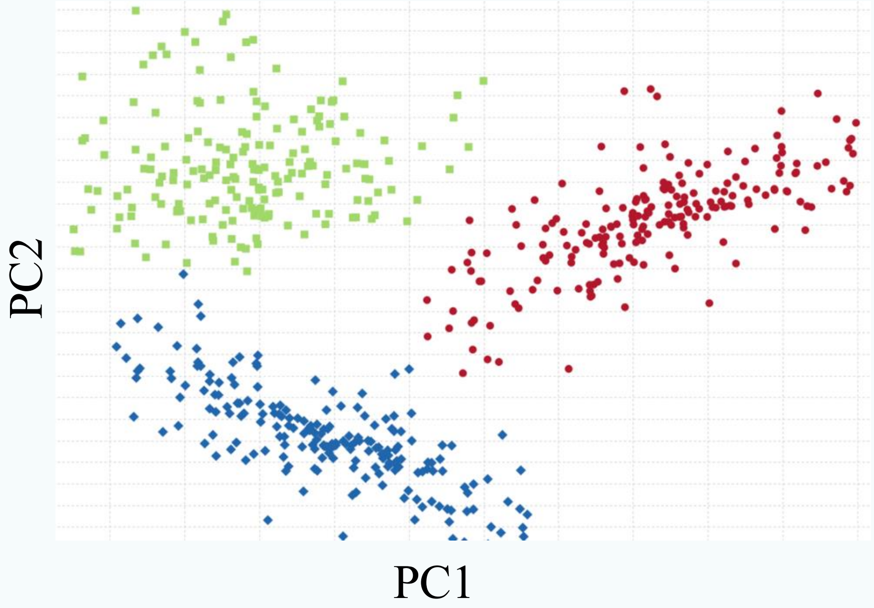


Connect K nearest neighbors

KNN graph



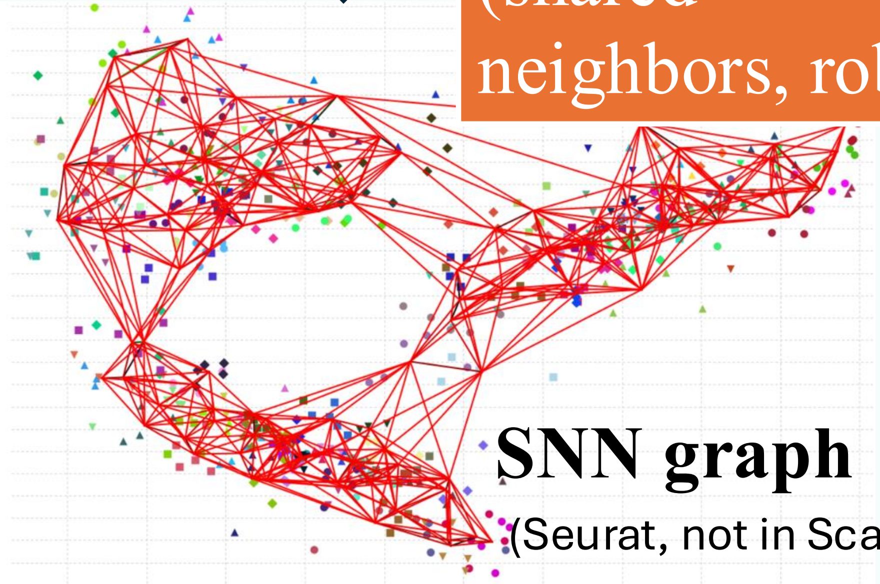
PCA space: on top n PCs



Connect K nearest neighbors

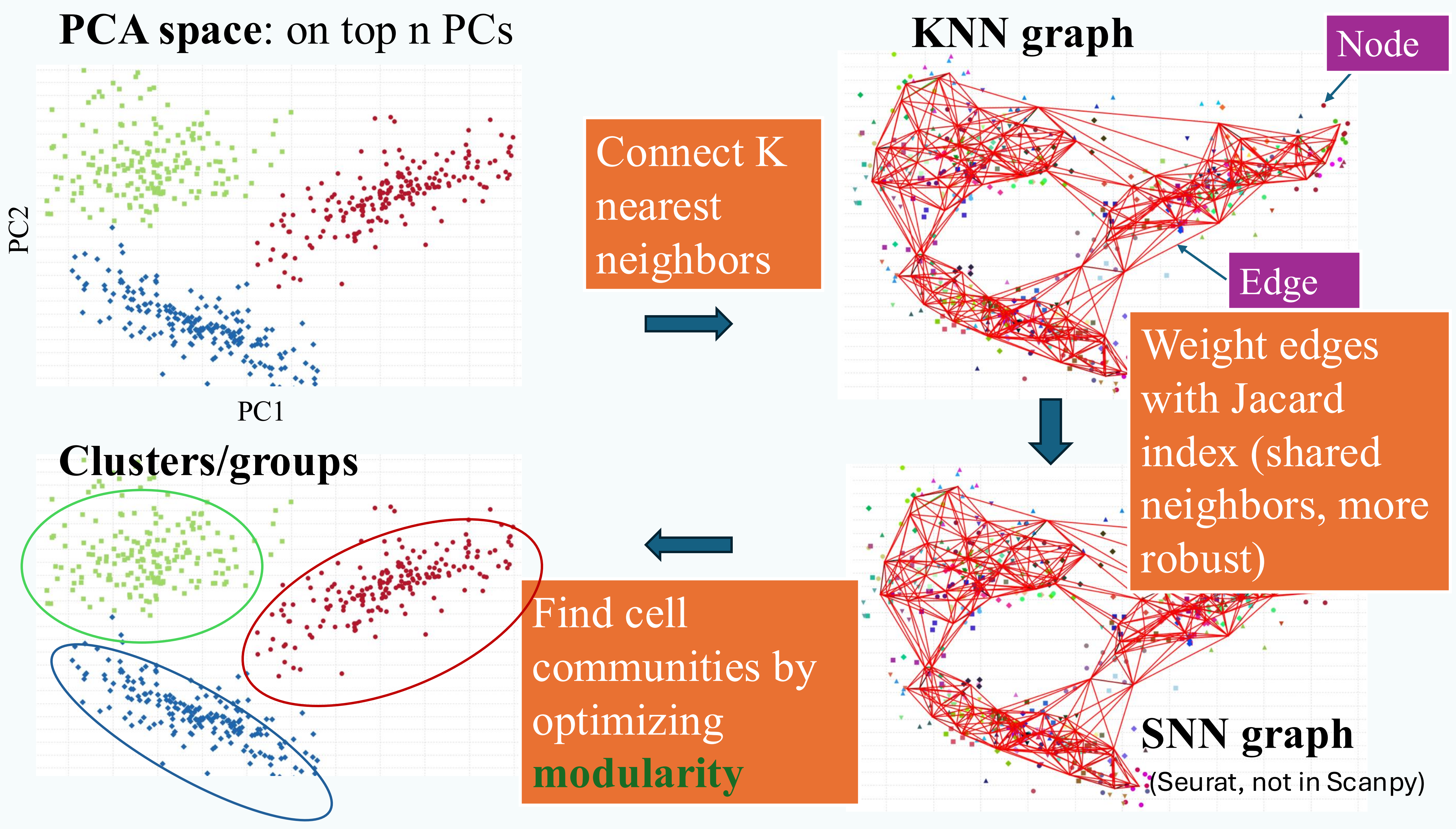
KNN graph

Weight edges with Jacard index (shared neighbors, robust)



SNN graph

(Seurat, not in Scanpy)


PCA space: on top n PCs
PC2
PC1
KNN graph
Node
Edge
Connect K nearest neighbors
Weight edges with Jacard index (shared neighbors, more robust)
SNN graph
(Seurat, not in Scanpy)
Find cell communities by optimizing modularity
Clusters/groups

Modularity optimization

Maximize the difference between the observed and the expected edges within clusters

$$\mathcal{H} = \frac{1}{2m} \sum_c \Big(\underbrace{e_c}_{\text{Observed edges within cluster c}} - \gamma \underbrace{\frac{K_c^2}{2m}}_{\text{Expected edges within cluster c}} \Big)$$

Resolution: higher, more clusters

Observed edges within cluster c

Expected edges within cluster c

Modularity measures how well a graph decomposes into groups of cells that are densely connected internally but sparsely connected between groups.

Tragg et al. 2019 Scientific reports

Modularity optimization

Maximize the difference between the observed edges within clusters and the expected edges

$$\mathcal{H} = \frac{1}{2m} \sum_c \left(\underbrace{e_c}_{\text{Observed edges within cluster c}} - \gamma \underbrace{\frac{K_c^2}{2m}}_{\text{Expected edges within cluster c}} \right)$$

Resolution: higher, more clusters

Observed edges within cluster c

Expected edges within cluster c

Modularity optimization algorithms:

Louvain

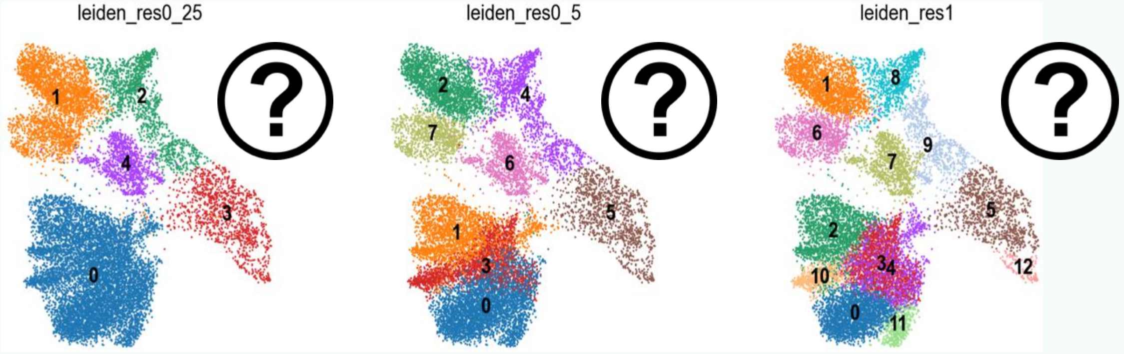
Leiden

Leiden: an improved version of Louvain

Modularity measures how well a graph decomposes into groups of cells that are densely connected internally but sparsely connected between groups.

Tragg et al. 2019 Scientific reports

Clustering resolution, quality evaluation

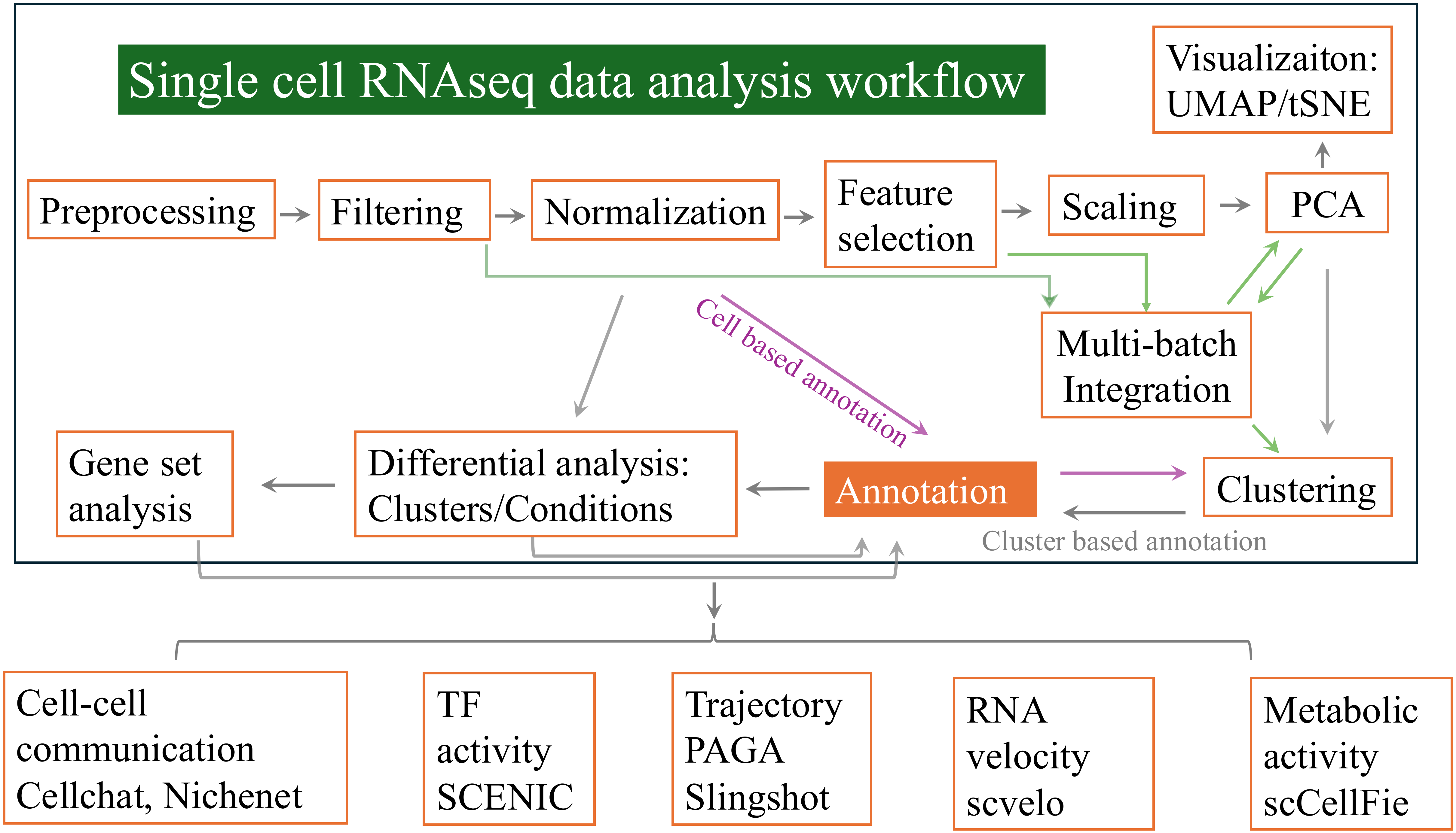


- **Resolution too low**: rare cell types merge with larger clusters, subtypes merge together
- **Resolution too high**: one celltype split to multiple clusters

- Measures
  - **Silhouette scores**
  - Calinksi-Harabasz index
  - Normalized mutual information
  - Variation of information
  - Graph-sensitive indices

- Cell-level automatic annotation before clustering
- The cluster-specific differentially expressed genes
- Biology

https://support.sas.com/resources/papers/proceedings19/3409-2019.pdf



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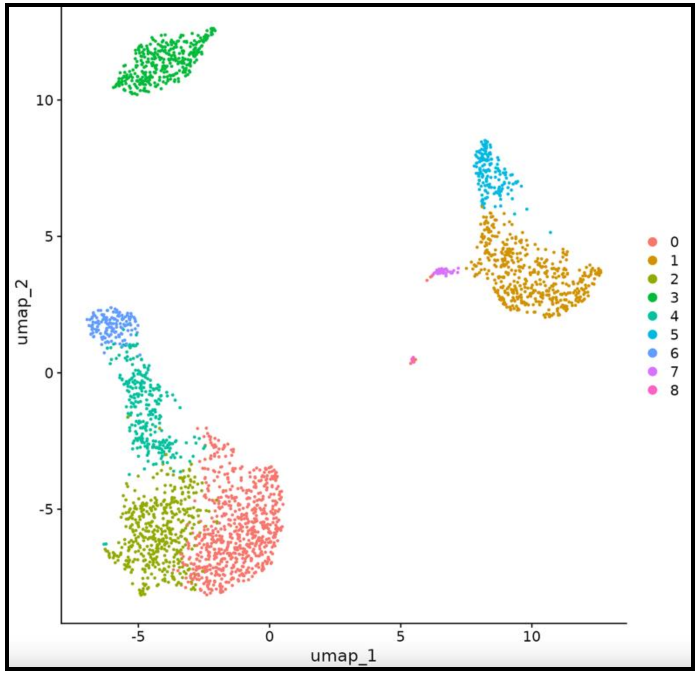
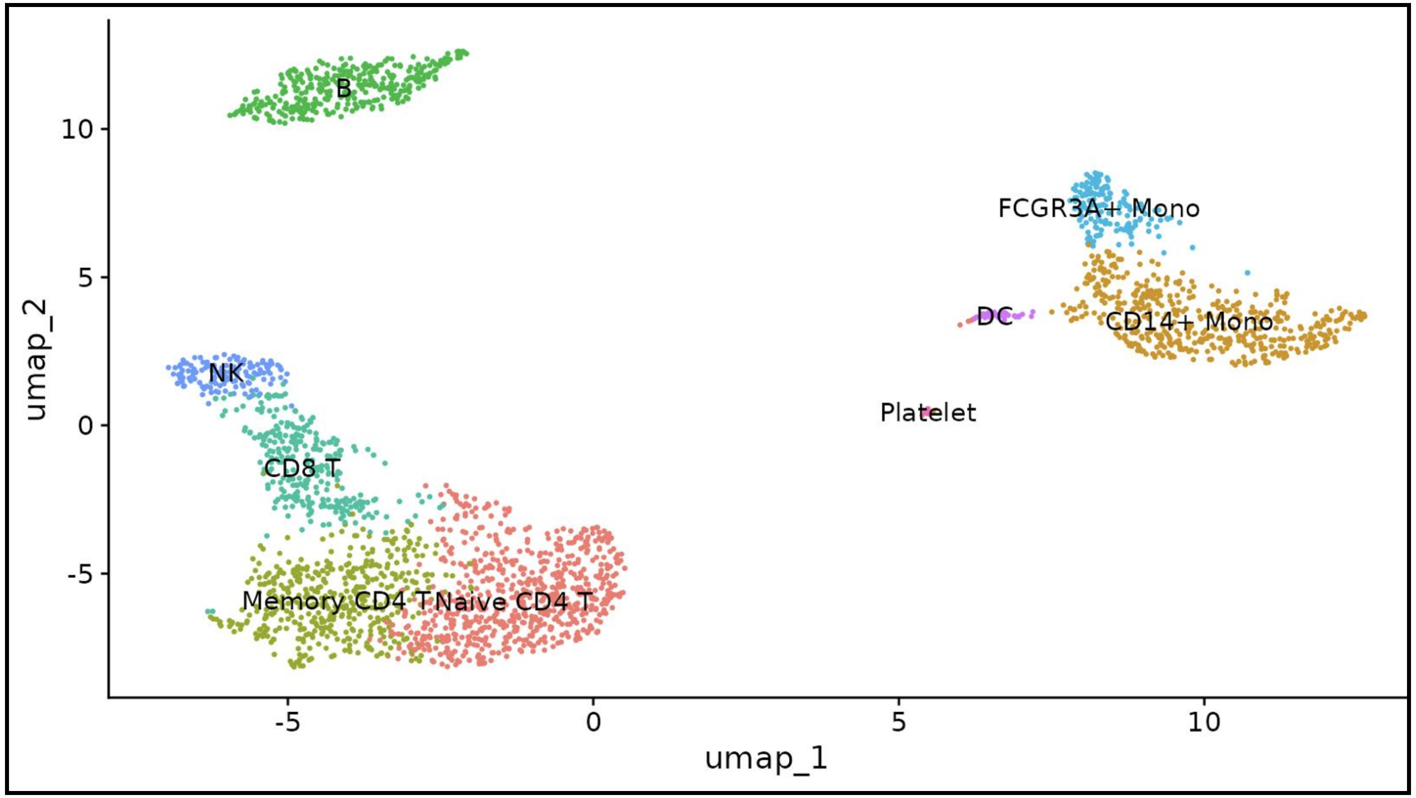
Cell annotation

Cell annotation: the process of labeling cells in your data with biol. meaningful celltypes



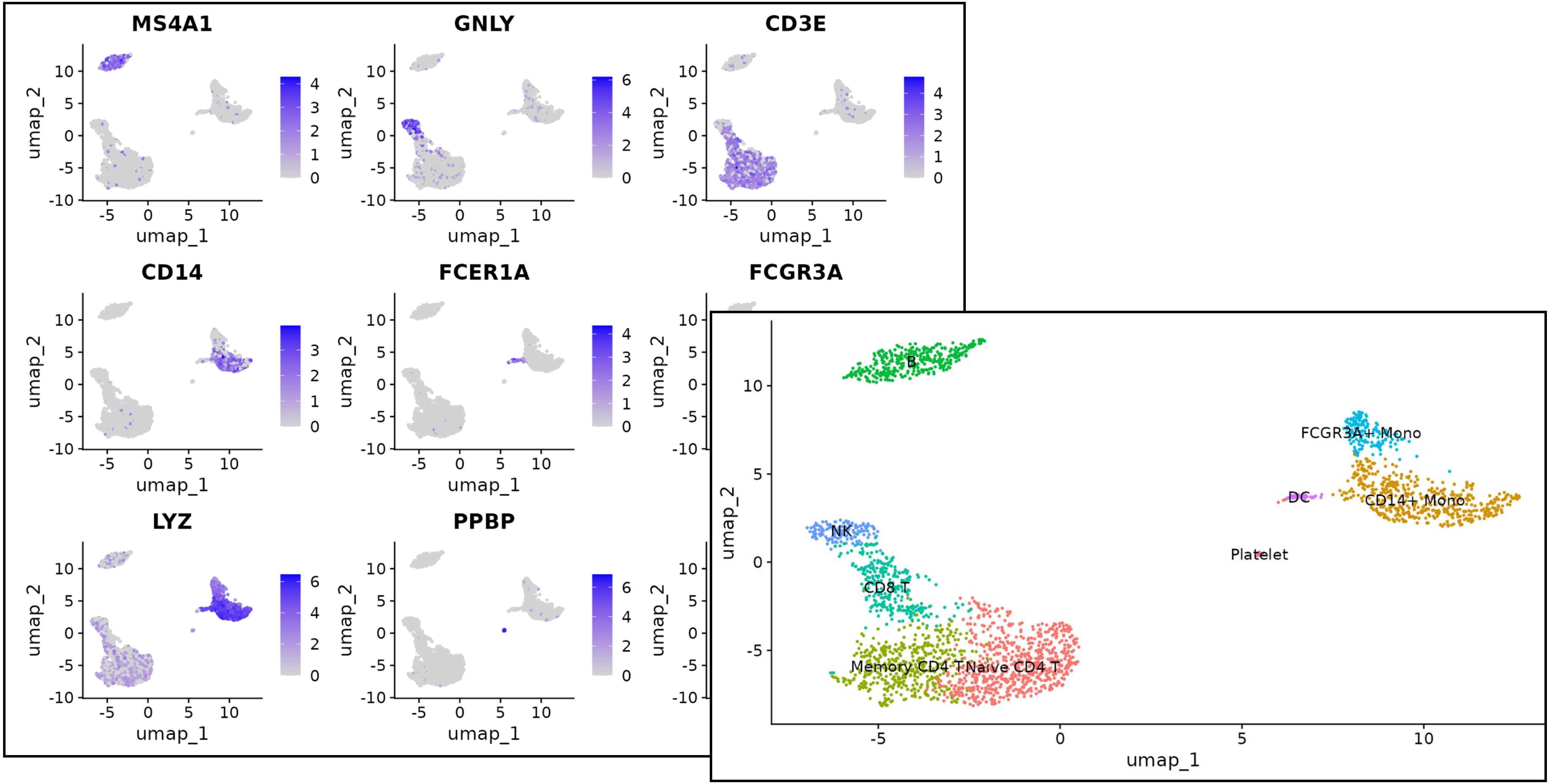


Celltype

A cellular phenotype:

- ➢Identifiable based on expression of specific markers (i.e. proteins or gene transcripts)
- ➢Often linked to specific functions
- ➢E.g. ?

Manual annotation



Manual annotation

- Based on a small set of marker genes (*better from RNAseq paper, not from protein study*) known to be associated with a particular cell type.

- **Two angles:**
  1. Celltype marker genes → their expression in your dataset
  2. Cluster-specific genes → known celltype marker genes

Manual annotation

Potential Caveats	Recommendation
Slow, labour-intensive.	**Whenever possible, begin with automatic annotation to determine general cell labels.**
Subjective.	**Work with an expert; consider multiple cell-type conclusions.**
Cell types not distinguishable by a single marker.	Use multiple markers for each cell type.
Known markers not distinguishing cell types.	Curate larger lists of markers from the literature, additional experiments or experts.
Conflicting marker gene sets between sources.	Select a marker gene set that best represents the biological signal being looked for in the data (e.g. if looking for cell subtypes, use more extensive gene sets than what is used for general cell-type annotation).

Clarke et a. 2021 Nature

❖ PROS:
- Transparent,
- Novel celltypes

→

Automatic annotation

- Fast
- Reproducible

Automatic annotation: comput. algorithm + prior biological knowledge

- Quickly identify known celltypes
- Highlight unknown celltypes for further investigation (new, subtypes)

Cell-level annotation

- Ideal (will not miss diff. among cells)
- Low seq. depth: zeros (expressed, not detected), sparse, not enough information single cells

Cell cluster (group) annotation

- More accurate, robust
- Clustering quality
- Compare multiple methods

Clarke et a. 2021 Nature protocols

Automatic annotation: comput. algorithm + prior biological knowledge

- Quickly identify known celltypes
- Highlight unknown celltypes for further investigation (new, subtypes)

Marker based

- **Known marker genes sources**: SCSig, PanglaoDB, CellMarker, DISCO, literature
- Conflict between sources
- Tools: AUCell, SCINA, GSEA/GSVA

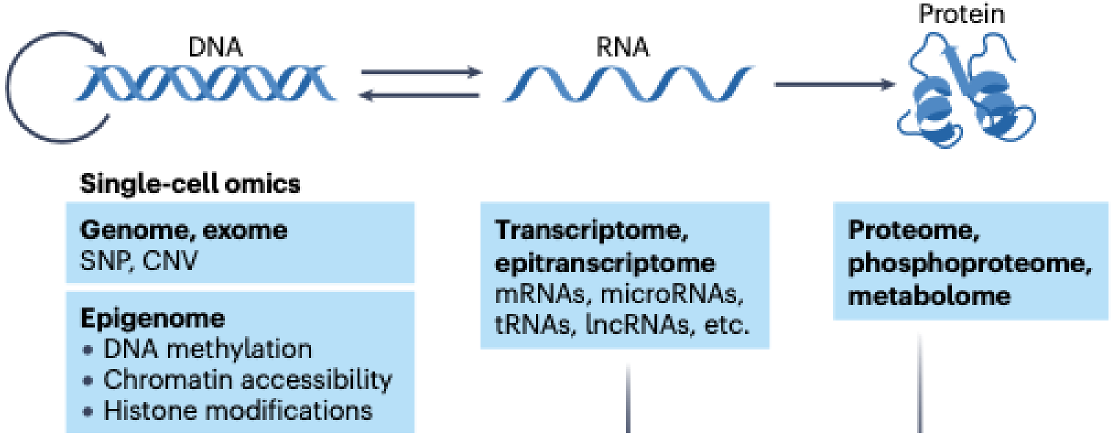
Reference dataset based

- **Query data**: your data, to be annotated
- **Reference data**: similar, expert-annotated dataset
- **Reference sources**: GEO, Single Cell Expression Atlas, cell atlas projects, DISCO, CellTypist, Azimuth, literature, Gut (https://www.gutcellatlas.org/pangi.html), cell ontology (https://cell-ontology.github.io/)
- Tools: Scmap, singleCellNet, singleR, Seurat

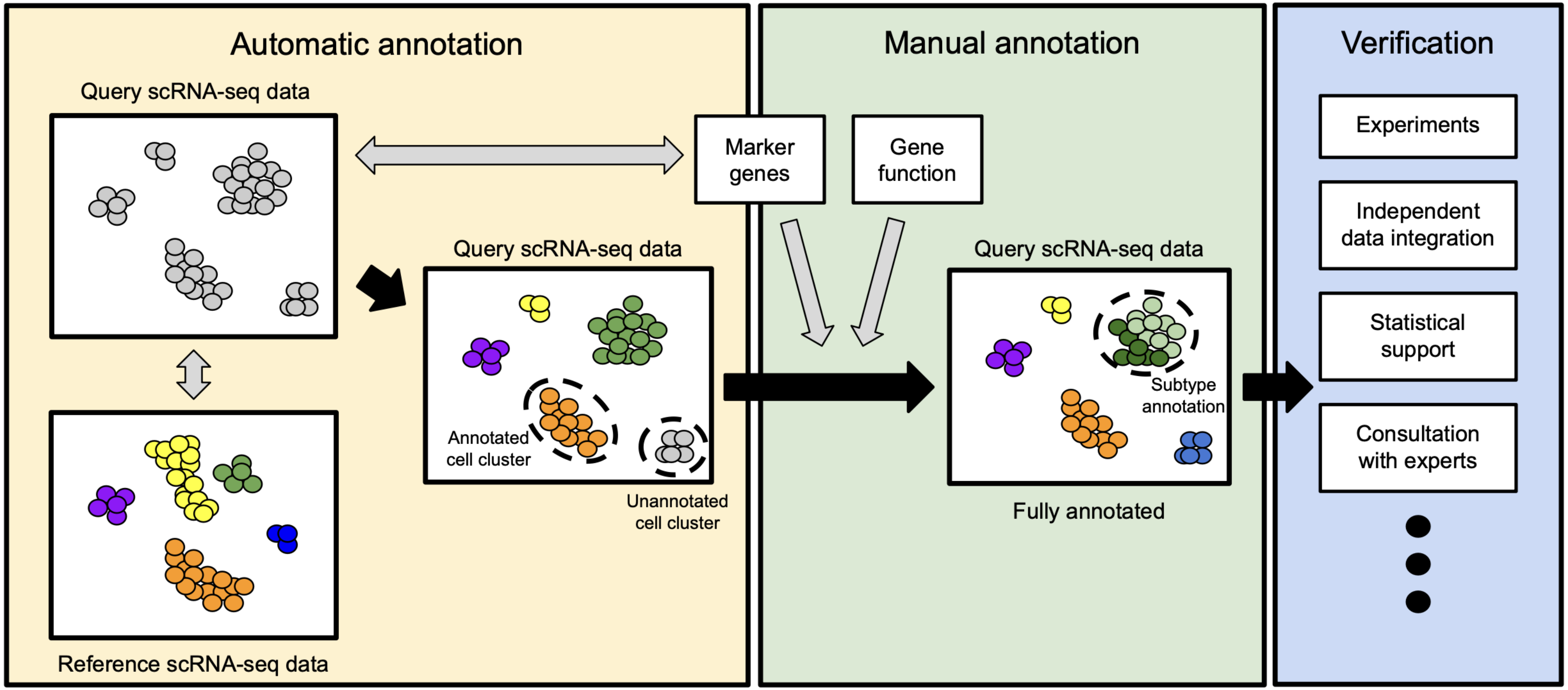
Clarke et a. 2021 Nature protocols

Annotation verification

- Statistical methods (e.g. Posterior inclusion probabilities, Chung 2020)
- Experimental validation:
  - Funcitonal assays (e.g. cytokine secretion, proliferative capacity),
  - Fluoresence in situ hybridization of source tissue sampels,
  - Validate marker gene expression in a larger number of samples
- Complementary genomic methods
  - Proteome, metabolome
  - Genome (SNP)
  - Epigenome
  - Spatial



Recommended workflow for annotation

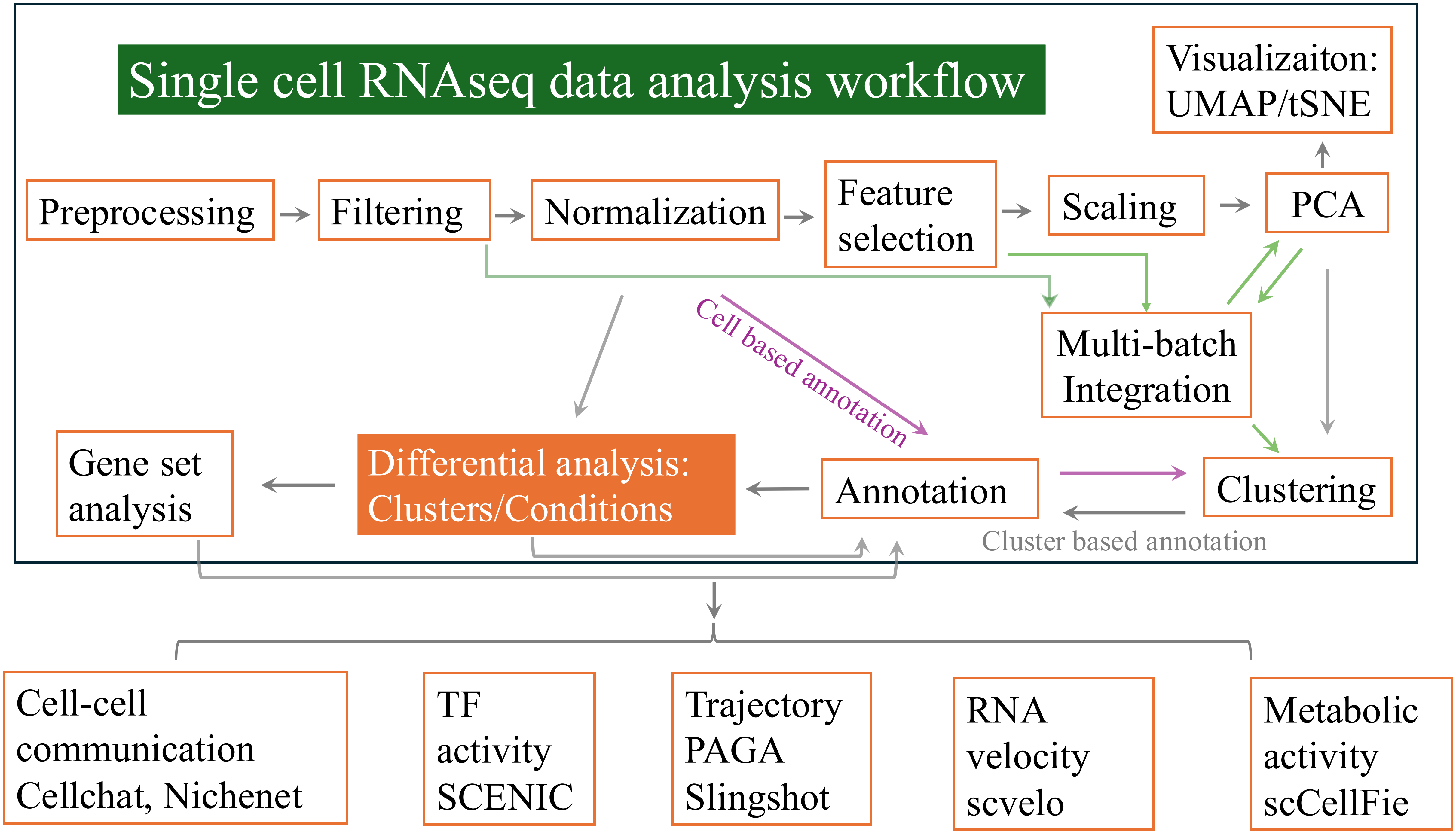


Clarke et al. 2021, Nature protocols

Pair discussion:

- How can we use cell-level automatic annotation to choose clustering resolution?

Differential expression analysis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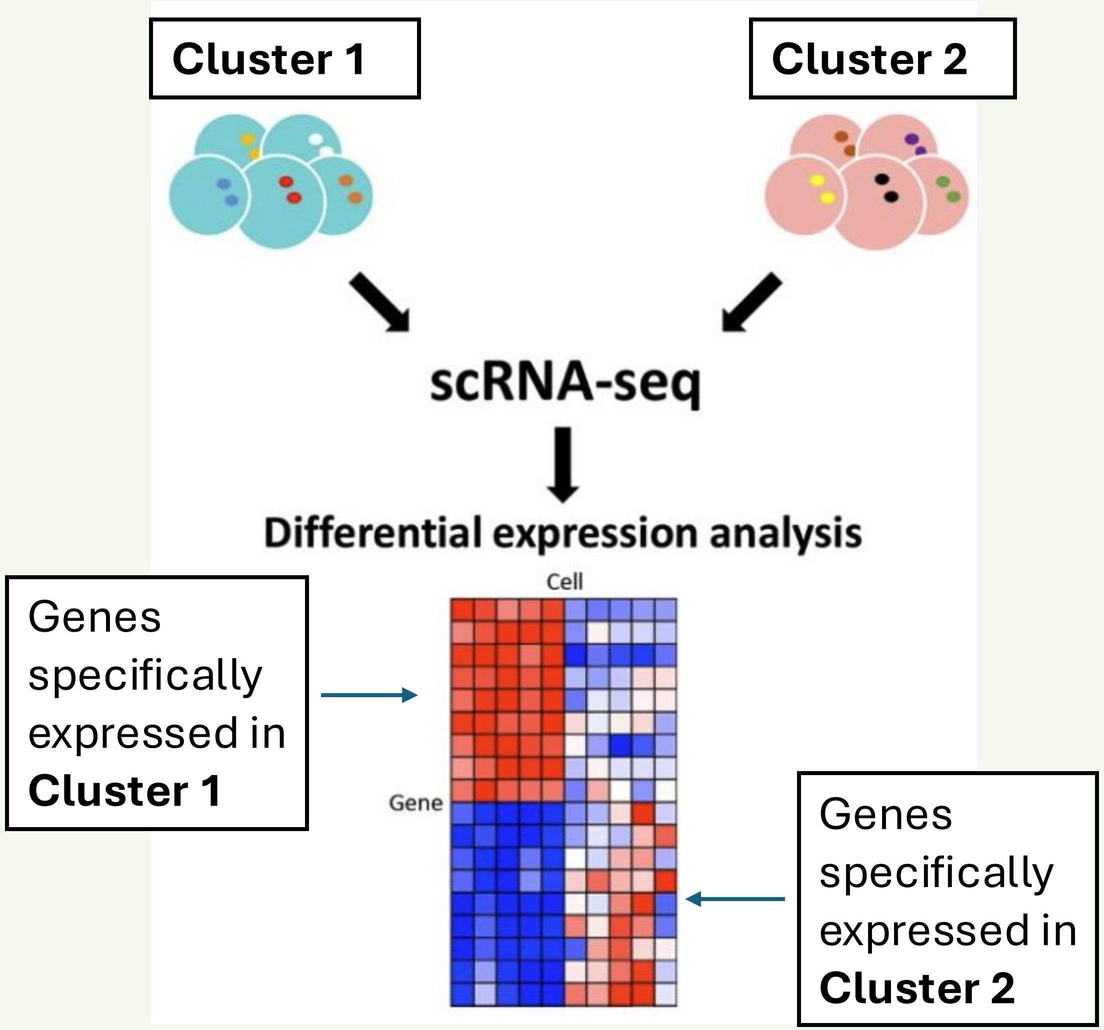
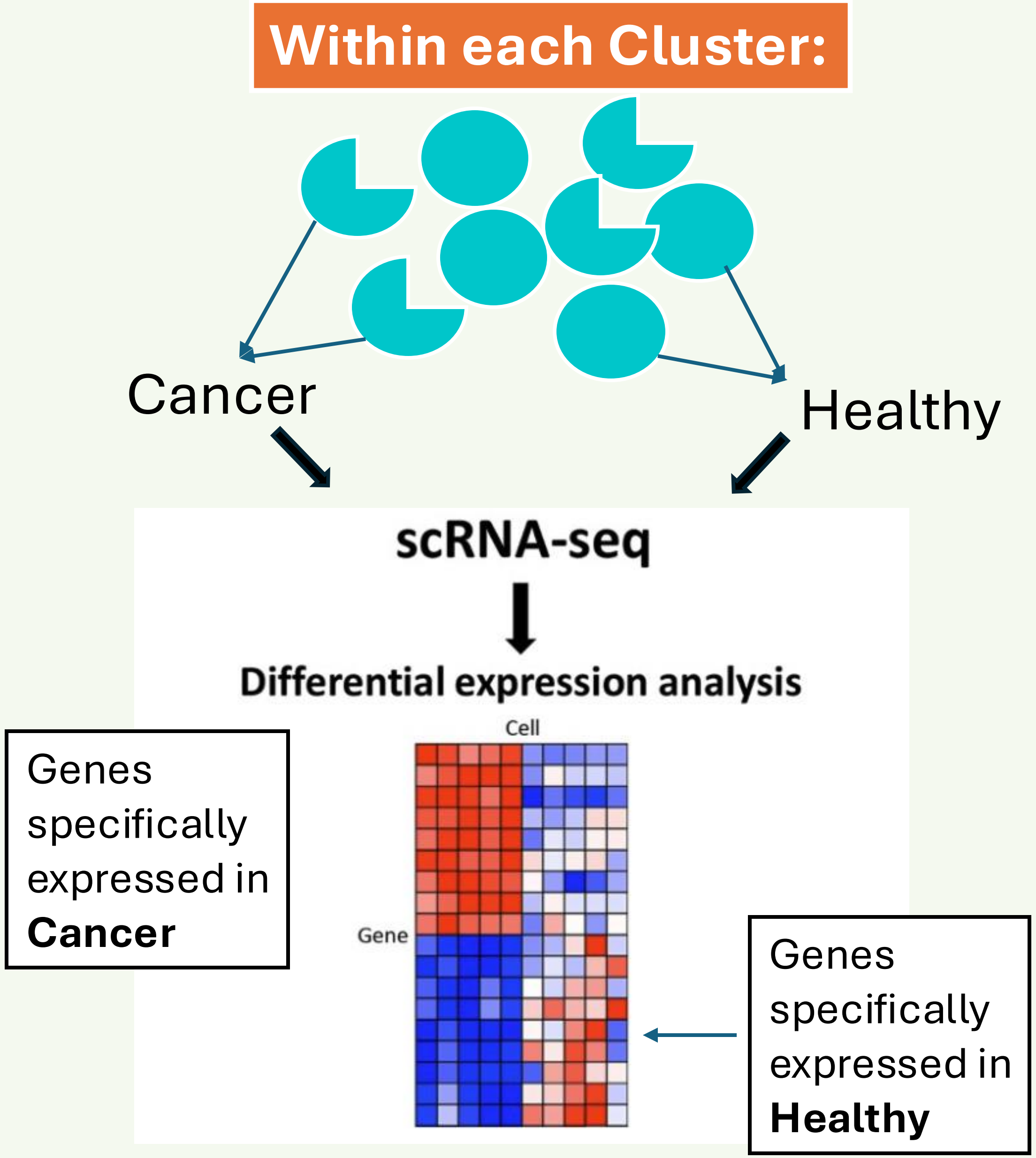
Between clusters



Between conditions



Methods for scRNAseq differential analysis:

Single cell RNAseq data:

- Count data (possion, NB, Guassian…)
- Noisy
- dropout
- Bi-modal
- Multi-modal

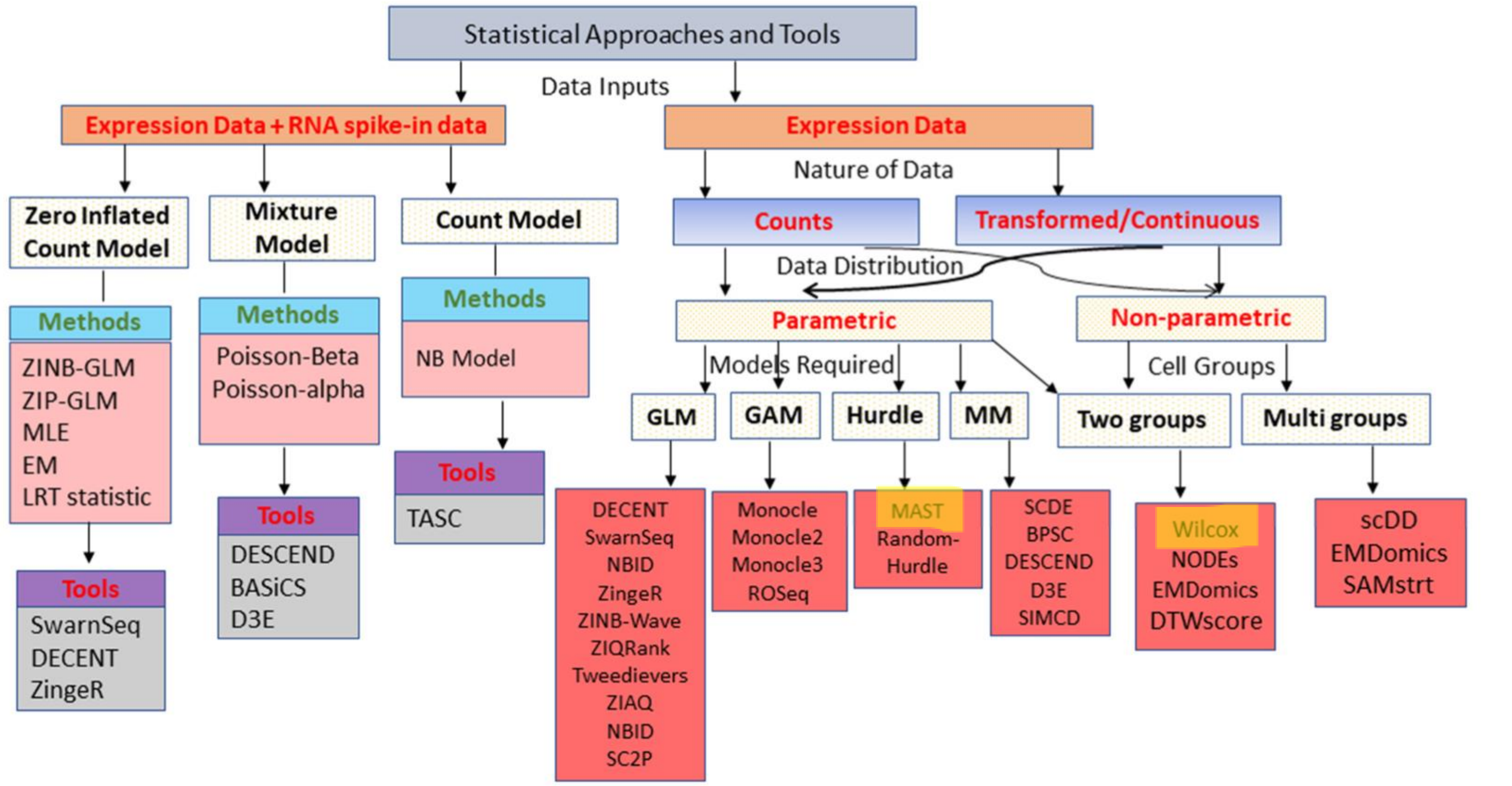
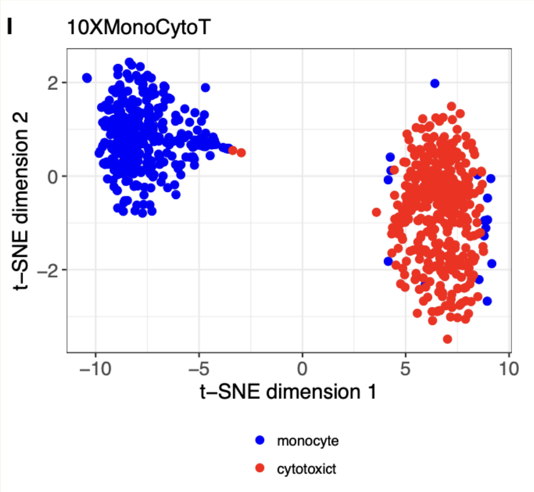
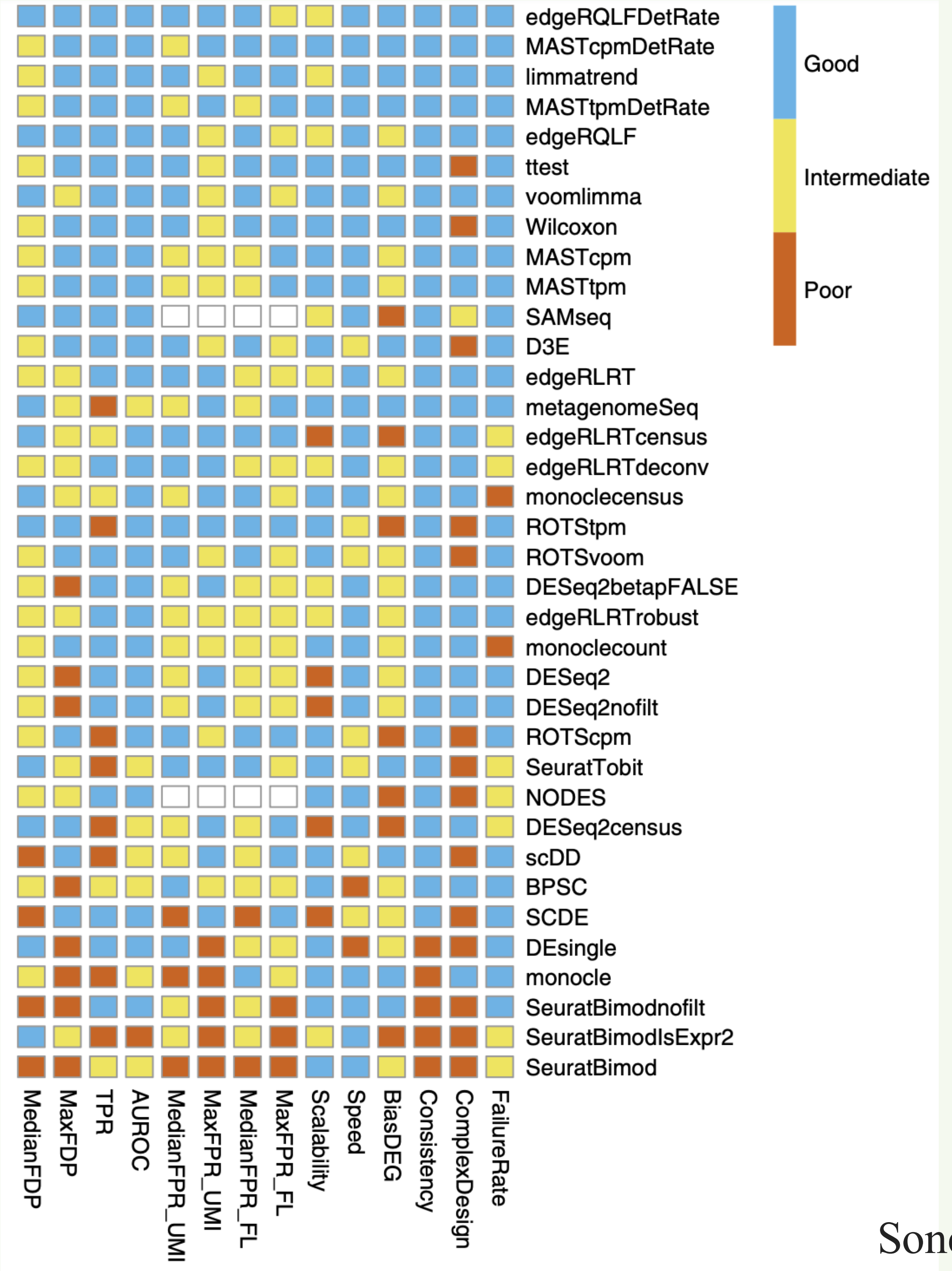


Figure 2. Classification of available statistical approaches and tools used for DEA in single-cell studies. Classification of the approaches is conducted based on the requirement of input data, data

Das et al. 2022 Entropy



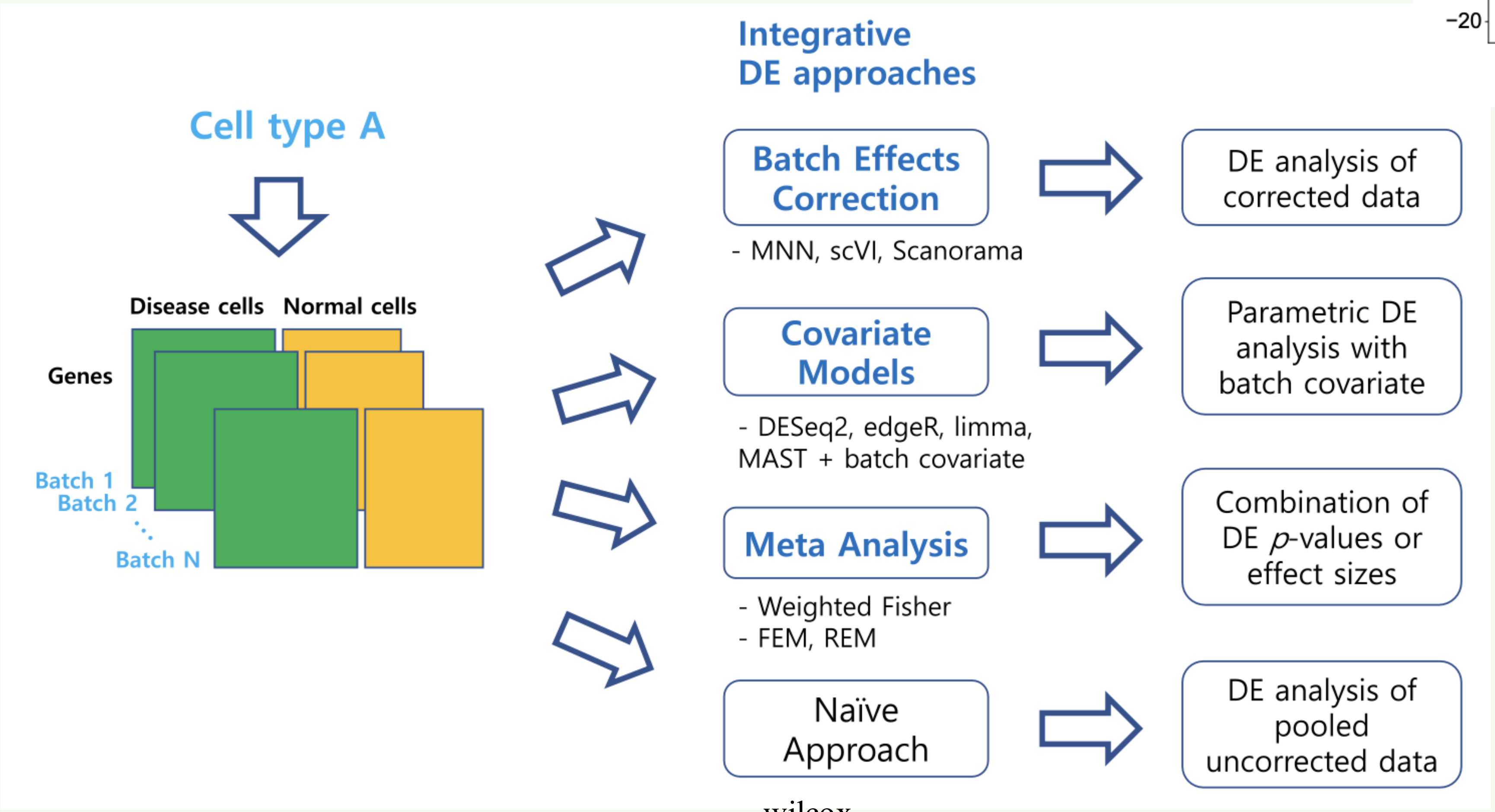
For simple comparison between 2 groups of cells:
No batches…



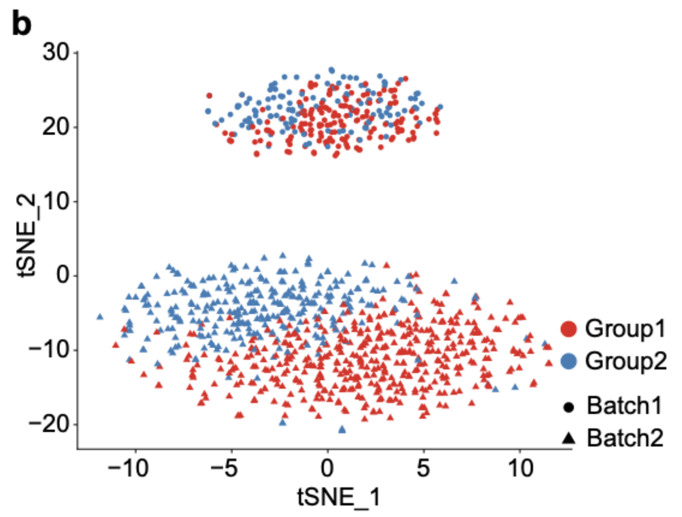
- ❖ t-test and Wilcoxon work well
- ❖ bulk RNA-seq analysis methods (DESeq2, edgeR, Limma…) do not generally perform worse than those specially developed for scRNA-seq

Soneson&Robinson 2018 *Nat Methods*

When batches added in



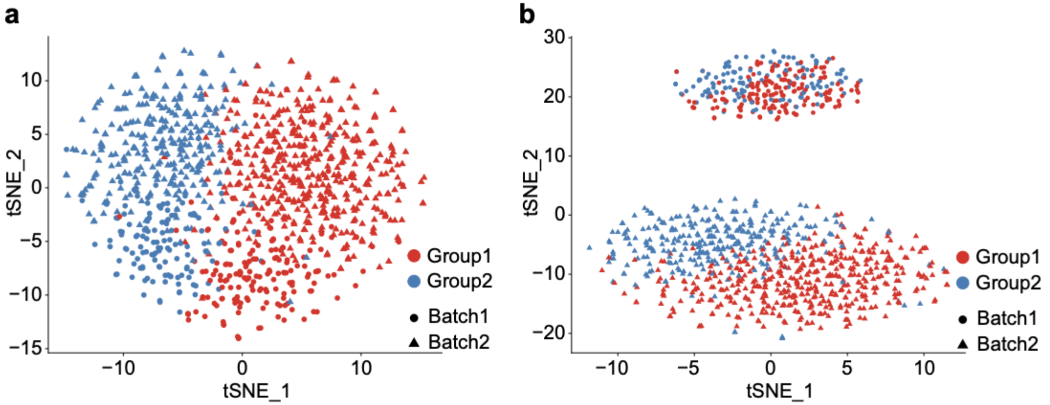
- wilcox



Nguyen et al. 2023 *Nat Commun*

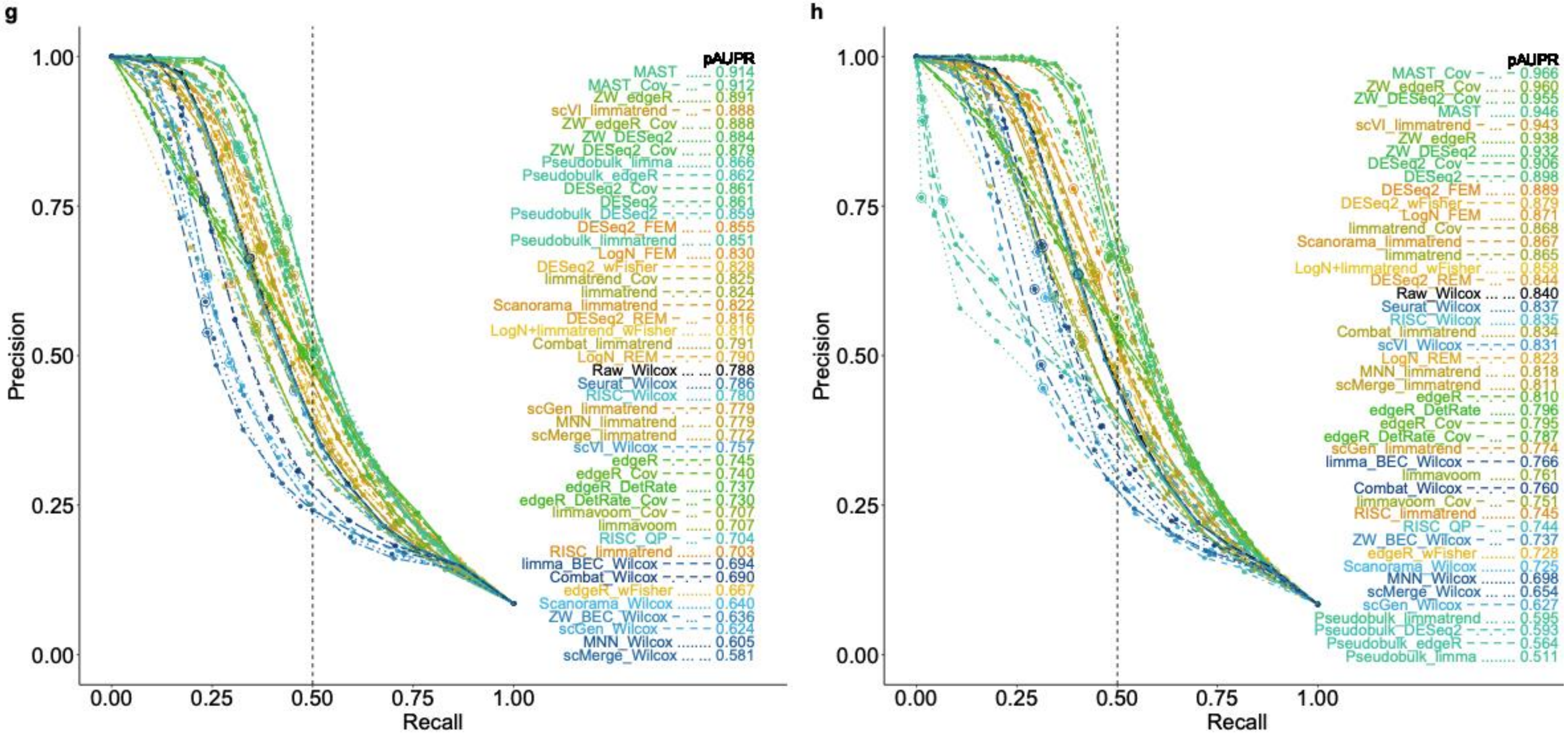
Nguyen et al. 2023 *Nat Commun*

- ❖ Intermediate depth (mean non-zero count of 77)
- ❖ High overall zero rate (> 80%)



Small batch effect

Large batch effect

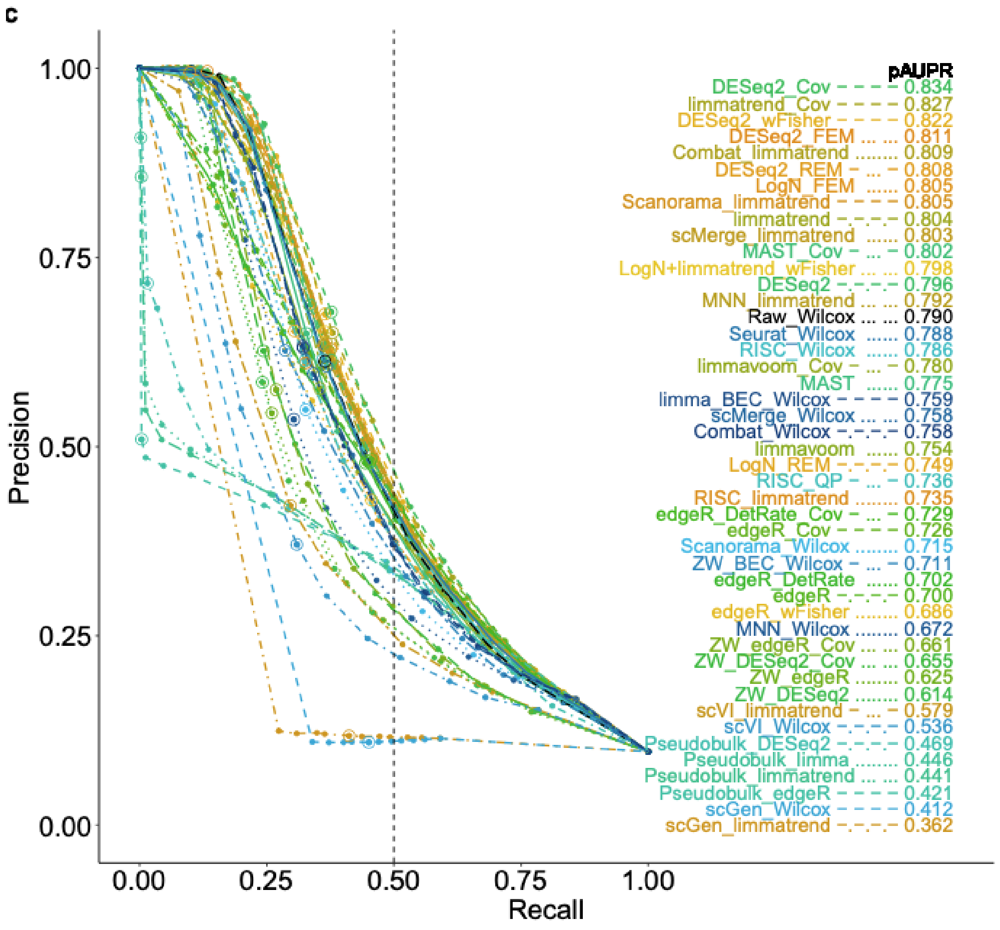


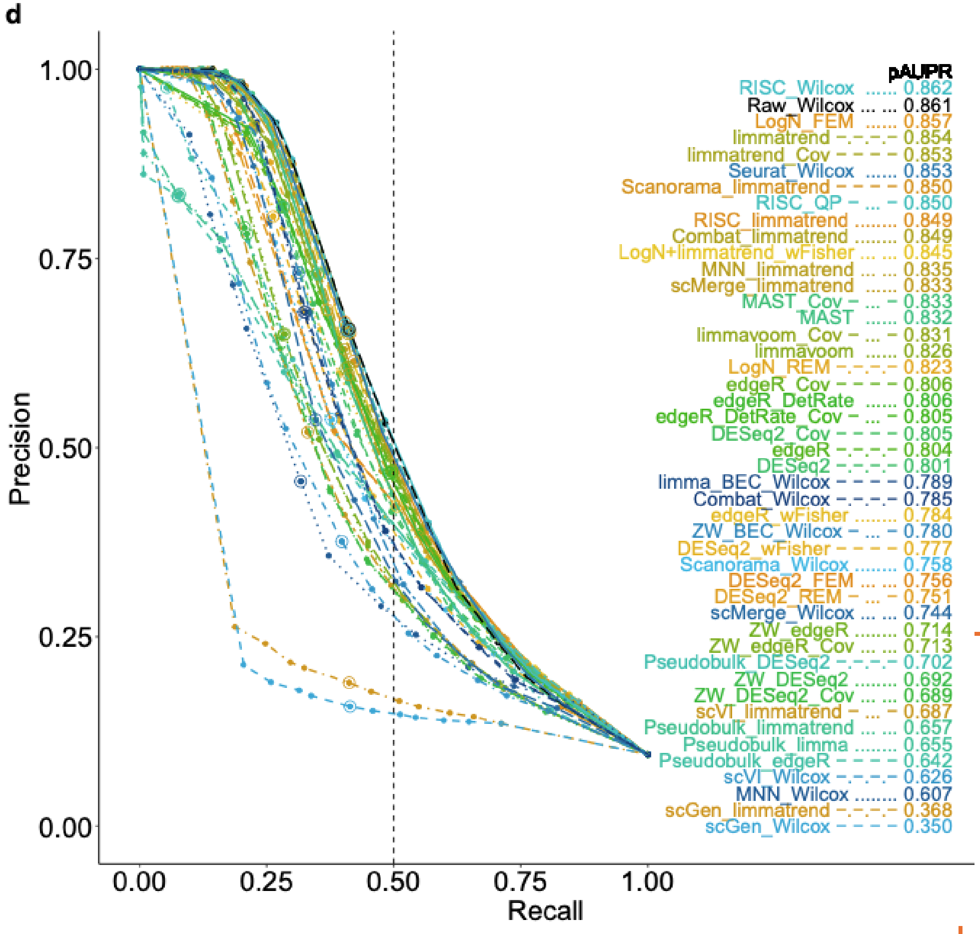
General performance
pAUPR: precision-recall curve {precision=TP/(TP+FP), Recall=TP/(TP+FN)}

- ❖ Low depth (mean non-zero count of 10 & 4/10X Genomics)
- ❖ Zero rate > 80%

Mean non-zero count of 10

Mean non-zero count of 4 (**10X**)





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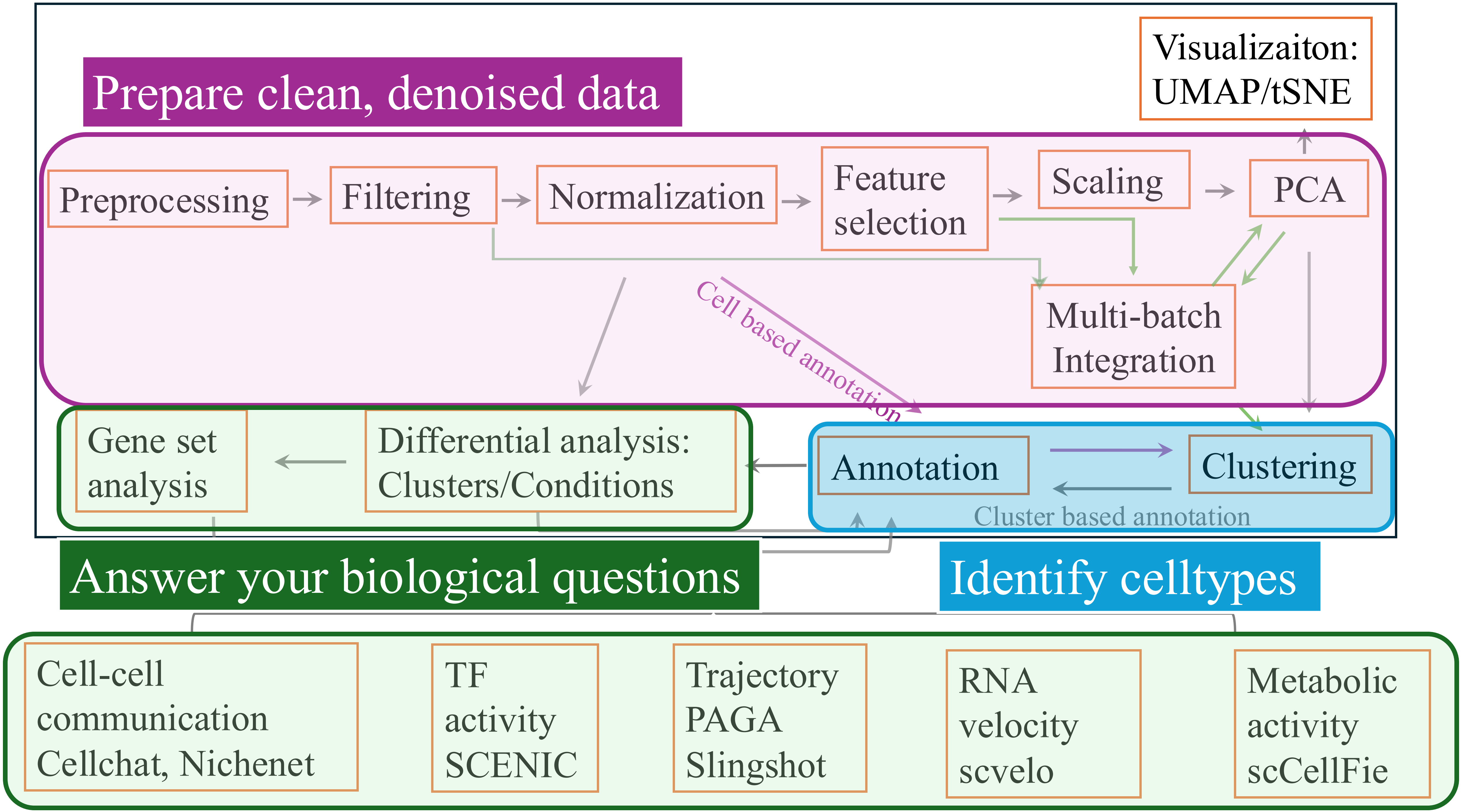
Nguyen et al. 2023 *Nat Commun*

Basic single cell data analysis summary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Pair discussion:

- Why and how do we filter the scRNAseq data?

Pair discussion:

- Why and how to do dimensional reduction?

Pair discussion:

- When do integration?

Pair discussion:

- Why do clustering?

References

- **Review:** Andrews, Tallulah S., et al. "Tutorial: guidelines for the computational analysis of single-cell RNA sequencing data." *Nature protocols* 16.1 (2021): 1-9.
- **SCTranform:** Hafemeister, Christoph, and Rahul Satija. "Normalization and variance stabilization of single-cell RNA-seq data using regularized negative binomial regression." *Genome biology*20.1 (2019): 296.
- **Integration**: Ryu, Yeonjae, et al. "Integration of single-cell RNA-seq datasets: a review of computational methods." *Molecules and cells* 46.2 (2023): 106-119.
- **Integration**: Luecken, Malte D., et al. "Benchmarking atlas-level data integration in single-cell genomics." *Nature methods* 19.1 (2022): 41-50.
- **Dimensional reduction**: Moon, Kevin R., et al. "Visualizing structure and transitions in high-dimensional biological data." *Nature biotechnology* 37.12 (2019): 1482-1492.
- **Clustering**: Traag, Vincent A., Ludo Waltman, and Nees Jan Van Eck. "From Louvain to Leiden: guaranteeing well-connected communities." *Scientific reports* 9.1 (2019): 1-12.
- **Clustering evaluation**: https://support.sas.com/resources/papers/proceedings19/3409-2019.pdf
- **Differential analysis:** Soneson, Charlotte, and Mark D. Robinson. "Bias, robustness and scalability in single-cell differential expression analysis." *Nature methods* 15.4 (2018): 255-261.
- **Differential analysis**: Nguyen, Hai CT, et al. "Benchmarking integration of single-cell differential expression." *Nature Communications* 14.1 (2023): 1570.
- **Cell annotaiton**: Clarke, Zoe A., et al. "Tutorial: guidelines for annotating single-cell transcriptomic maps using automated and manual methods." *Nature protocols* 16.6 (2021): 2749-2764.